STM32F405 QUADCOPTER FLIGHT CONTROLLER

PCB Design Portfolio

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4-Layer KiCad Design | JLCPCB Fabrication | Betaflight Compatible

Microcontroller	STM32F405RGTx — ARM Cortex-M4 @ 168 MHz
Input Voltage	4S–6S LiPo (14.8V–25.2V)
Power Architecture	TPS54360DDA Buck → 5V | SPX3819 LDO → 3.3V
IMU	ICM-42688-P (SPI1, 6-DOF, 32kHz ODR)
Sensors	BMP388 Baro | QMC5883L Mag | SAM-M10Q GPS
OSD	AT7456E with 27 MHz crystal
PCB	4-Layer, 74×108 mm, HASL, FR4 TG135
Motor Outputs	4× DShot (TIM1 CH1-3, TIM8 CH4)

Table of Contents

1. Project Overview

This portfolio documents the complete design of a custom quadcopter flight controller built from scratch using KiCad EDA. The board targets Betaflight firmware compatibility and is designed for 4S–6S LiPo battery systems (14.8V–25.2V nominal). Every component value, footprint, and routing decision was made with explicit engineering justification grounded in datasheets, signal integrity principles, and power electronics theory. The board was fabricated by JLCPCB as a 4-layer PCB and partially assembled via their SMT service, with through-hole and large SMD connectors hand-soldered.

The design goal was to produce a professional-grade flight controller equivalent to commercial offerings such as the Matek F405 or Holybro Kakute F4, while applying first-principles engineering at every stage. This includes full datasheet-driven component selection, calculated feedback networks, analog power isolation, differential pair routing, and a deliberate 4-layer stackup strategy to minimize EMI and maximize signal integrity.

2. System Architecture

The flight controller is organized around three functional domains: power conversion, the main processing core, and the sensor/peripheral subsystem. Battery voltage enters through two XT30 connectors (J9 for ESC power pass-through, J10 for FC input), is stepped down by a synchronous buck converter to 5V, then regulated further by an LDO to 3.3V for digital logic and sensors. The analog supply (VDDA) for the STM32's ADC is further filtered from the 3.3V digital rail using a ferrite bead and capacitor network.

Block	Component	Interface	Rail
Main MCU	STM32F405RGTx	—	3.3V
Buck Converter	TPS54360DDA	—	VBAT → 5V
LDO	SPX3819M5-L-3-3	—	5V → 3.3V
IMU	ICM-42688-P	SPI1 (PA5/6/7, CS:PC4)	3.3V
Barometer	BMP388	SPI3 (PB3/4/5, CS:PC0)	3.3V
Magnetometer	QMC5883L	I2C2 (PB10/11)	3.3V
GPS	SAM-M10Q-00B	UART3 (PB0/1)	3.3V
OSD	AT7456E	SPI2 (PC3/PB13/PC2, CS:PC5)	5V
ESD Protection	USBLC6-2SC6	USB D+/D–	—
Motor Outputs	4× JST-SH	TIM1 CH1-3, TIM8 CH4	5V
Receiver	JST-SH 4-pin (J5)	UART1 (PA9/10)	5V
VTX	JST-SH 4-pin (J4)	UART2 (PA2)	5V

3. Power Architecture

The power system must handle a wide input range (4S LiPo: 14.8V nominal, 16.8V full charge; 6S: 22.2V nominal, 25.2V full charge) and deliver clean, stable power to a mixed analog-digital load. A two-stage approach was chosen: a high-efficiency switching buck converter to handle the large voltage step-down, followed by a low-noise LDO for the final digital rail, with additional LC filtering for the sensitive analog domain.

3.1 TPS54360DDA Buck Converter — VBAT → 5V @ 3.5A

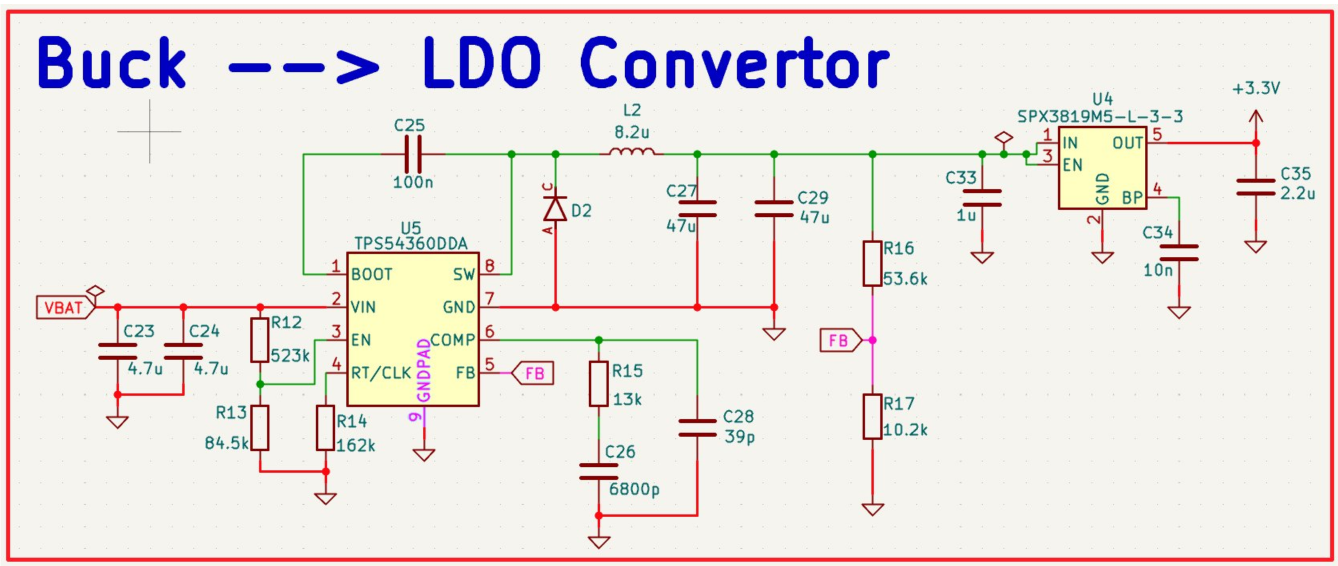

Figure 1: Complete power section schematic — TPS54360DDA buck converter and SPX3819 LDO

The Texas Instruments TPS54360DDA was selected for its wide input voltage range (4.5V–60V), 3.5A continuous output capability, internal high-side MOSFET, and availability through JLCPCB assembly. The switching frequency was set to 600 kHz by the RT resistor R14 (162 kΩ), calculated directly from the TPS54360 datasheet equation: R_T = (1/f_sw − 100ns) / (10.05 × 10■^{12}). Higher switching frequency allows a smaller inductor at the cost of slightly higher switching losses — 600 kHz is the recommended sweet spot for this IC balancing efficiency and component size.

Output Voltage Feedback Network (R15, R16, R17):

R15 (upper, FB to VOUT) = 13 kΩ

R16 (lower, FB to GND) — sets output voltage via voltage divider

Target output: 5.0V. Equation: V_OUT = V_REF × (1 + R_upper/R_lower), V_REF = 0.8V

Calculated: R16 = 10.2 kΩ, R17 = 53.6 kΩ — verified against datasheet

UVLO (Undervoltage Lockout) — R12, R13:

R12 = 523 kΩ (upper), R13 = 84.5 kΩ (lower)

Sets UVLO start threshold ~14.5V and stop ~13V, preventing operation below 4S minimum

Protects battery from deep discharge and ensures stable startup

Compensation Network — R15, C26, C28:

R15 = 13 kΩ, C26 = 6800 pF, C28 = 39 pF

Type II compensation network stabilizes the control loop across all load conditions

C26 sets the zero frequency, C28 adds a high-frequency pole to attenuate switching noise

Values calculated per TPS54360 datasheet Section 9.2 compensation procedure

Inductor — L2 (8.2 µH):

L = (V_IN – V_OUT) / (f_sw × ΔI_L) — calculated for 30% peak-to-peak current ripple

At V_IN=25.2V, V_OUT=5V, f_sw=600kHz, I_OUT=3A: L ≈ 8.2 µH

RS0630-8R2MS selected: 7.3×7.3mm power inductor, I_sat > 4A, low DCR

Catch Diode — D2 (B560C-13-F):

Schottky diode, 60V/5A rated, DO-214AB (SMA-C) package

Placed as close as possible to the SW pin — this is the critical hot node

Low forward voltage (0.52V typ) minimizes conduction losses during freewheeling phase

Input Capacitors — C23, C24 (4.7 µF × 2, 1210):

Placed immediately at VBAT/VIN pins to decouple high-frequency switching transients

1210 package chosen for high voltage rating (≥50V) and low ESR at switching frequency

Two caps in parallel halves effective ESL and ESR

Output Capacitors — C27, C29 (47 µF × 2, 1210):

Large bulk capacitance on output to minimize ripple and support load transients

Placed at output node, away from IC side, to complete the output current loop locally

Bootstrap Cap — C25 (100 nF):

Connected between BOOT and SW pins per datasheet

Provides gate drive voltage for the internal high-side MOSFET

Must be placed as close to BOOT pin as possible

3.2 SPX3819M5-L-3-3 LDO — 5V → 3.3V

The MaxLinear SPX3819M5-L-3-3 LDO was chosen for its low dropout voltage (340 mV at 500 mA), low quiescent current (90 µA), and built-in enable pin and thermal/overcurrent protection. It takes the clean 5V buck output and produces 3.3V for the STM32 and all digital sensors. The total 3.3V load was calculated from datasheets: STM32F405 (~50 mA active), ICM-42688-P (3.1 mA), BMP388 (0.7 mA), QMC5883L (2.5 mA), SAM-M10Q (15 mA typ), plus decoupling overhead, totaling approximately 99 mA — well within the 500 mA rating.

Input cap C33 = 1 µF (stability, placed at IN pin)

Output caps C34 (10 nF) + C35 (2.2 µF) — output filter, improves PSRR and transient response

EN pin connected via R16/R17 voltage divider from buck output — enables LDO only when 5V is stable

BP pin bypassed with 10 nF (C34) per datasheet requirement

3.3 VDDA Analog Power Filter

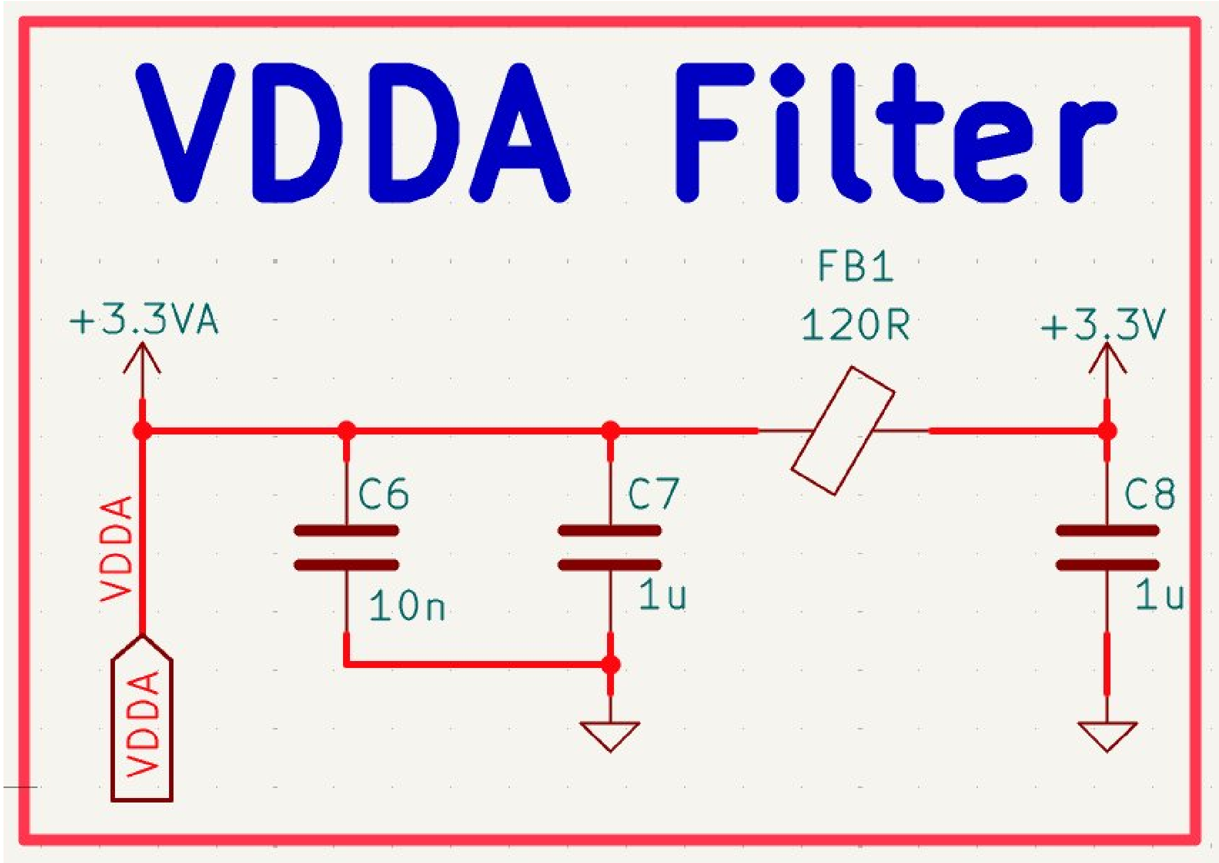

Figure 2: VDDA analog supply filter — FB1 ferrite bead with C6/C7/C8 capacitor network

The STM32F405 has a dedicated VDDA pin for its ADC, DAC, and analog comparator circuitry. Feeding VDDA directly from the digital 3.3V rail would couple switching noise and digital ground bounce directly into the ADC reference, degrading measurement accuracy. A ferrite bead filter was therefore implemented between the digital 3.3V rail and VDDA.

FB1 — 120 Ω @ 100 MHz, 0402:

Sunlord GZ1005D121TF ferrite bead acts as a low-pass filter at high frequencies

Passes DC and low-frequency signals (negligible DC resistance) but attenuates switching noise above ~10 MHz

120 Ω impedance at 100 MHz chosen per STM32F4 reference manual recommendation

Filter Capacitors:

C6 = 10 nF (0402): high-frequency bypass, targets self-resonance near 100 MHz

C7 = 1 µF (0805): mid-frequency bulk decoupling, handles lower-frequency noise

C8 = 1 µF (0805): post-ferrite output stabilization on VDDA side

The combination creates a cascaded LC filter providing >40 dB attenuation of switching noise above 10 MHz

4. STM32F405 Microcontroller

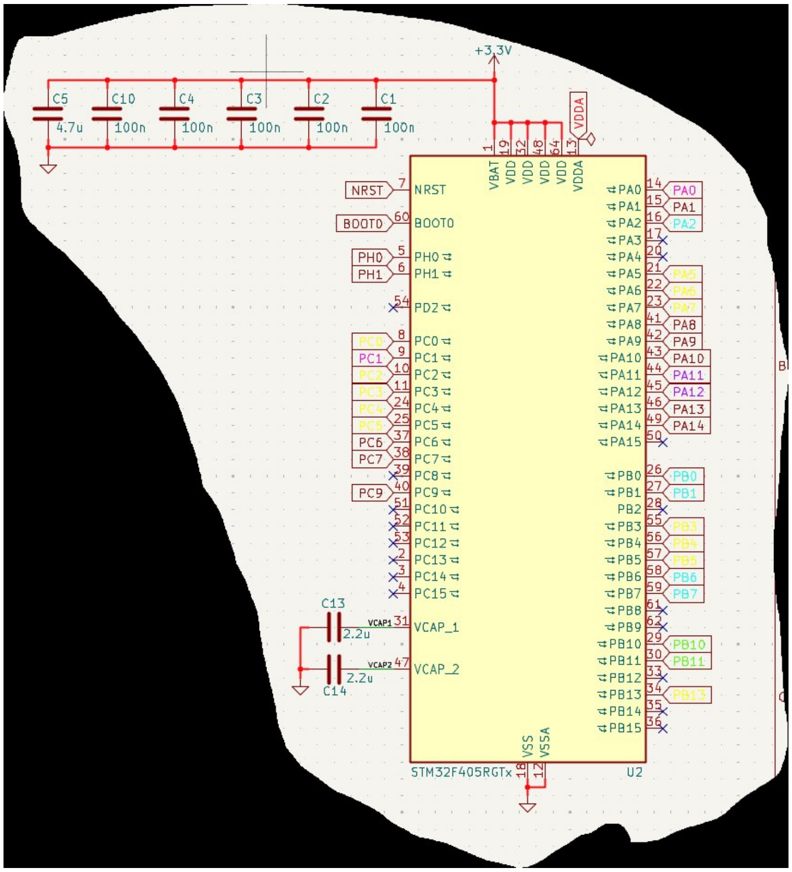

Figure 3: STM32F405RGTx schematic — power pins, decoupling network, VCAP capacitors

The STM32F405RGTx is an ARM Cortex-M4F microcontroller running at up to 168 MHz with an FPU. It was chosen over the STM32F411 specifically to gain additional UART peripherals, CAN bus support, and full alignment with existing Betaflight hardware targets. The LQFP-64 package was used for its 0.5 mm pitch solderable leads, allowing JLCPCB SMT assembly.

4.1 Power Decoupling Network

Per the STM32F405 datasheet Figure 17, each VDD pin must have its own 100 nF decoupling capacitor placed as close as possible to the pin. The STM32F405 has four VDD pins (pins 19, 48, 63, 64) and one VDDA pin. The decoupling strategy follows three principles:

- C1–C4 = 100 nF 0402 (X7R) — one per VDD pin, placed within 1–2 mm of each pin
- C5 = 4.7 µF 0805 — bulk bypass for the combined VDD network, handles low-frequency transients
- VDDA supplied through FB1 ferrite filter (see Section 3.3)

The 100 nF capacitors are sized to have their self-resonant frequency near the switching frequencies of concern (~100 MHz for digital noise). At resonance, impedance is minimized to ESR only (~10 mΩ), providing the lowest possible impedance path for transient return current. Using 0402 packages minimizes parasitic inductance compared to larger packages, keeping self-resonant frequency high. The 4.7 µF bulk cap provides charge reservoir for larger, slower transients such as peripherals switching on.

VCAP Pins (C13, C14 = 2.2 µF each):

- VCAP1 (pin 31) and VCAP2 (pin 47) connect to the internal voltage regulator output
- The STM32F405 has an internal 1.2V core voltage regulator requiring external capacitors on VCAP for stability
- 2.2 µF specified by STM32F405 datasheet — must not be omitted or the core regulator will oscillate
- Connected to GND, not 3.3V — these are regulator output caps, not bypass caps

4.2 Crystal Oscillator — 8 MHz

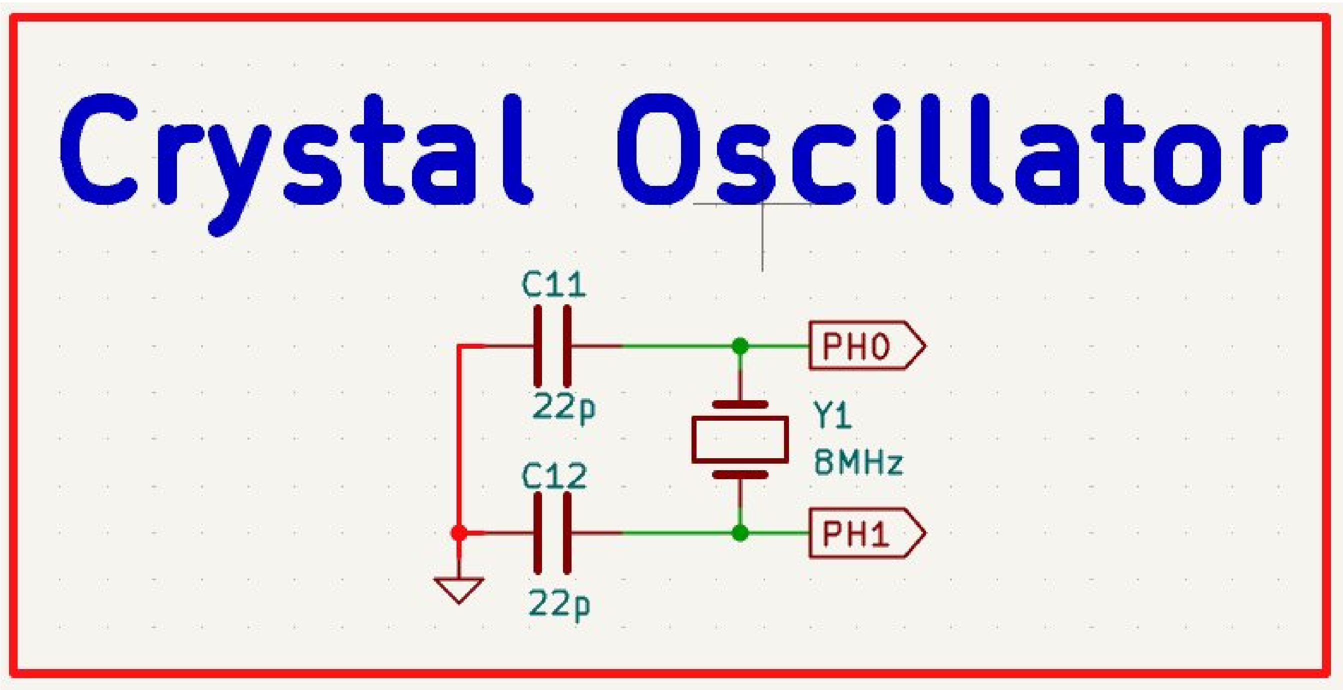

Figure 4: 8 MHz crystal oscillator circuit — Y1 with 22 pF load capacitors C11/C12

The STM32F405 uses a 4–26 MHz external crystal on the HSE (High Speed External) oscillator input at pins PH0 and PH1. An 8 MHz crystal was chosen because Betaflight targets the STM32F405 with an 8 MHz HSE, and the PLL is configured to multiply this to 168 MHz system clock. The crystal frequency must match the Betaflight firmware target configuration.

Load Capacitors — C11, C12 = 22 pF (C0G/NP0, 0402):

- Crystal oscillator circuits require load capacitors on each pin to ground
- Load capacitance C_L = (C11 × C12)/(C11 + C12) + C_stray ≈ 11 pF + C_stray
- Typical stray capacitance ~3–5 pF, giving effective C_L ≈ 14–16 pF
- C0G/NP0 dielectric chosen for temperature stability — critical for clock accuracy
- Traces from crystal pins to STM32 kept as short as possible (<5 mm) to minimize stray capacitance
- Traces surrounded by ground copper to shield from noise coupling

The 27 MHz crystal Y2 used for the AT7456E OSD follows the same design approach, as the OSD chip requires a separate clock independent of the MCU.

4.3 BOOT0 Configuration

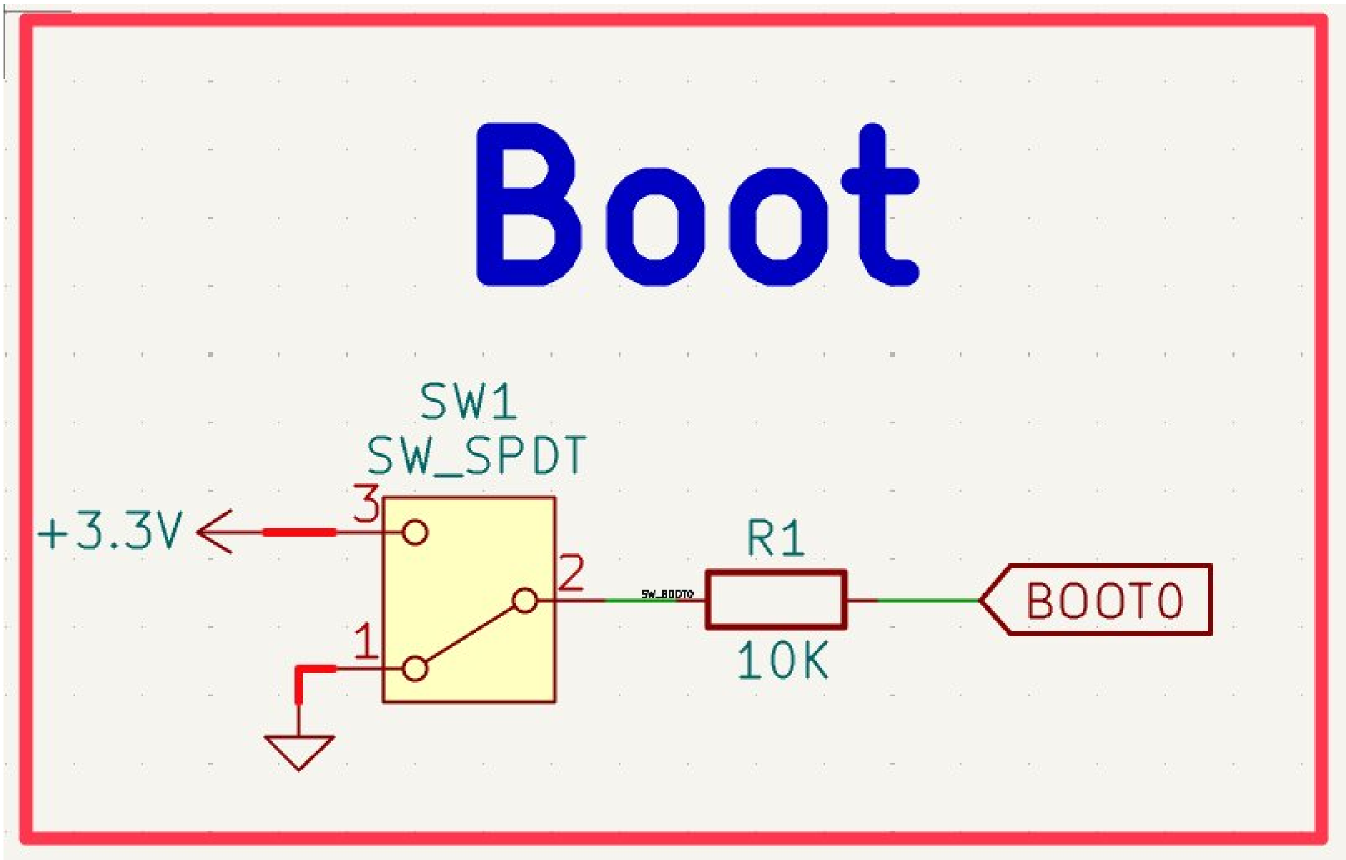

Figure 5: BOOT0 circuit — SPDT slide switch SW1 with 10 kΩ pull-down R1

The BOOT0 pin determines the boot mode of the STM32 on reset. When BOOT0 is low (GND), the STM32 boots from user flash — normal operation for Betaflight. When BOOT0 is pulled high (3.3V), the STM32 enters the internal bootloader, enabling USB DFU firmware flashing without a separate programmer.

SW1 = C&K; JS102011SAQN SPDT slide switch — selects between GND and 3.3V

R1 = 10 kΩ pull-down — ensures BOOT0 defaults to GND when switch is in run position

10 kΩ chosen to limit current while providing a strong enough logic low against noise

This eliminates the need for an ST-LINK for routine firmware updates — the user simply slides SW1, plugs in USB, and flashes via DFU, then slides back for normal operation.

4.4 NRST Reset Circuit

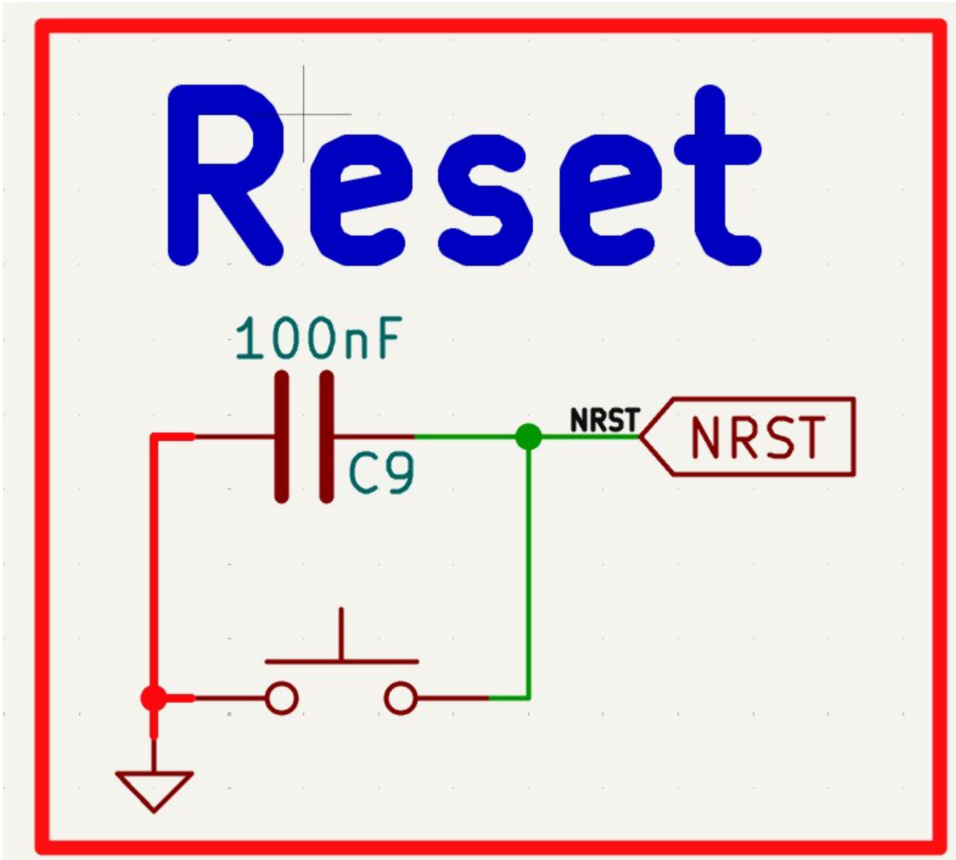

Figure 6: NRST reset circuit — momentary push button with 100 nF filter capacitor C9

The NRST pin is an active-low reset. A momentary push button (SW2) pulls NRST to GND when pressed, resetting the MCU. C9 (100 nF) is placed between NRST and GND as specified in the STM32 reference manual to filter out noise and ESD spikes on the reset line, preventing spurious resets in the electrically noisy drone environment.

- STM32 internal pull-up keeps NRST high during normal operation
- C9 = 100 nF creates an RC filter — time constant with internal ~40 kΩ pull-up is ~4 ms
- This debounces the mechanical switch and filters conducted EMI from ESCs

4.5 SWD Debug Interface

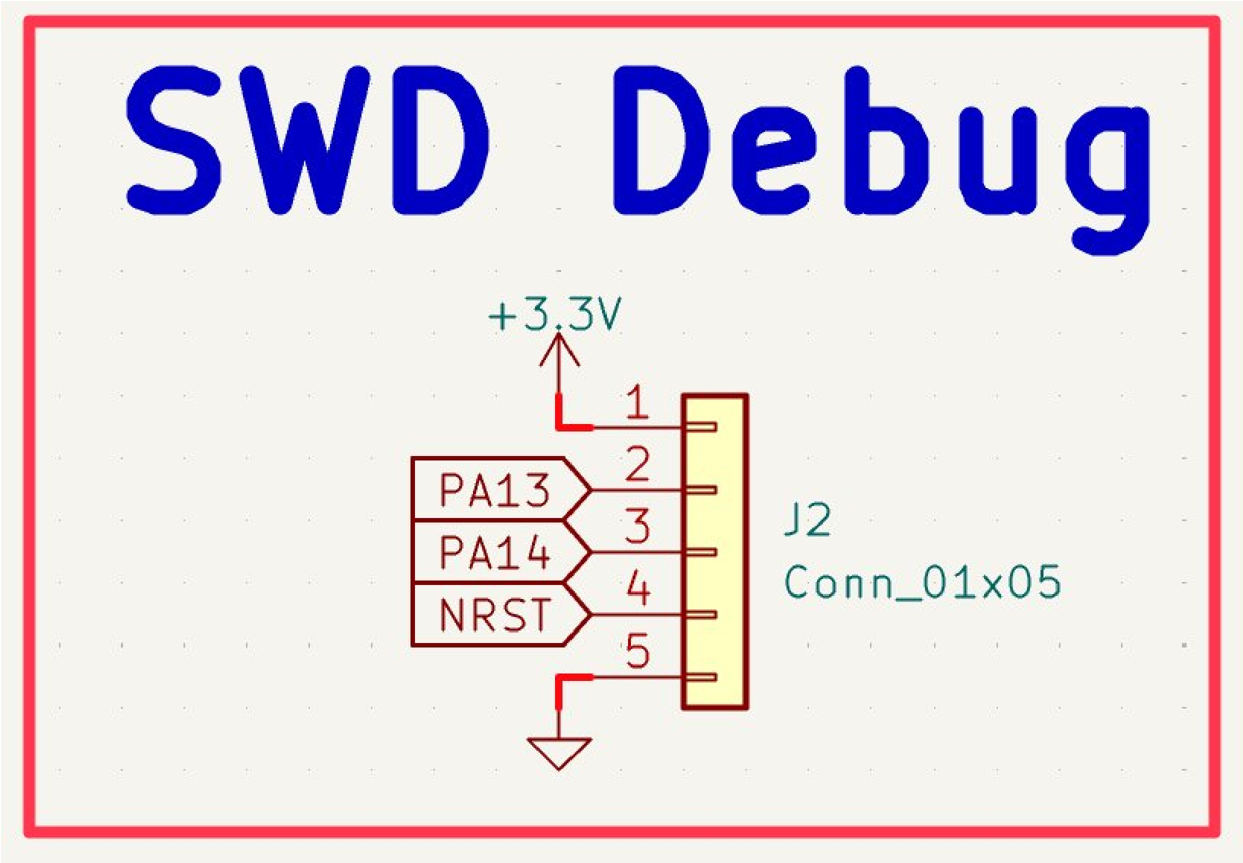

Figure 7: 5-pin SWD debug header J2 — SWDIO/PA13, SWCLK/PA14, NRST, 3.3V, GND

A 5-pin 2.54mm pitch through-hole header provides Serial Wire Debug (SWD) access. SWD requires only SWDIO (PA13) and SWCLK (PA14) plus ground for full debug capability, including breakpoints, register inspection, and memory access via an ST-LINK V2 probe. NRST and 3.3V are included for convenience — NRST allows the debugger to reset the target, and 3.3V allows the probe to power the board from the debug connector during development.

- PA13 = SWDIO — bidirectional data, 2.2 kΩ internal pull-up on STM32
- PA14 = SWCLK — clock from debug host, 40 kΩ internal pull-down on STM32
- Pinout matches standard ARM 5-pin SWD layout for ST-LINK compatibility

5. USB-C Interface

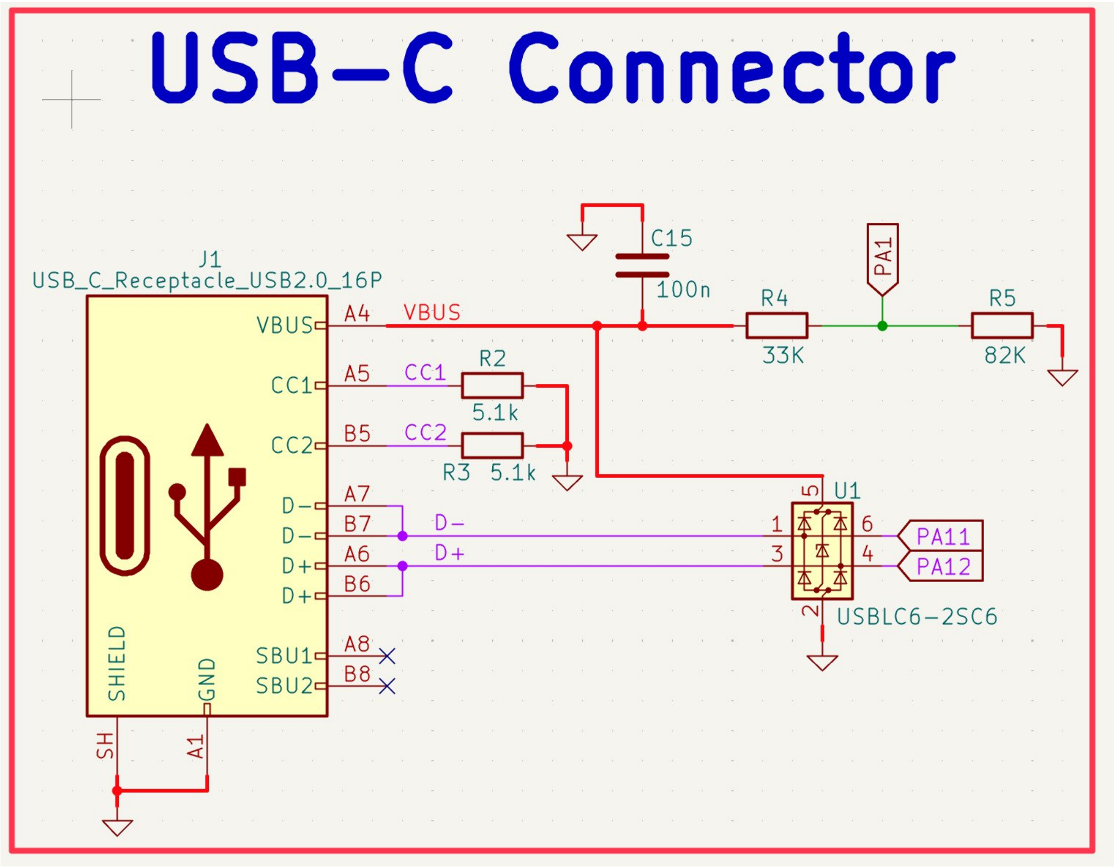

Figure 8: USB-C schematic — J1 connector, ESD protection U1, CC pull-downs, VBUS sense

USB-C is used for firmware flashing via DFU and Betaflight Configurator communication. The STM32F405 has a built-in USB OTG FS peripheral connected to PA11 (D–) and PA12 (D+). The design implements USB 2.0 Full Speed (12 Mbps) — sufficient for all configurator functions.

ESD Protection — U1 (USBLC6-2SC6, SOT-23-6):

ST USBLC6-2SC6 is specifically designed for USB 2.0 line protection

Ultra-low capacitance: 3.5 pF max — critical to not degrade USB signal integrity

Higher capacitance TVS diodes would round the USB signal edges and cause bit errors

Handles ±8 kV ESD strikes per IEC 61000-4-2 — protects STM32 USB pins from hot-plugging

Placed as close as possible to the USB connector, before any signal routing

CC Pull-Down Resistors — R2, R3 (5.1 kΩ):

USB-C CC (Configuration Channel) pins determine the role of each connector

5.1 kΩ pull-downs on CC1 and CC2 advertise this device as a USB device (sink)

This is required for USB-C to USB-A cables to work — the host sees a valid device

Without these resistors, USB-C hosts may not negotiate power or enumerate the device

Value per USB-C specification Table 4-25: Rd = 5.1 kΩ for default USB device role

VBUS Sense Divider — R4 (33 kΩ), R5 (82 kΩ):

VBUS (5V from USB host) is divided down to <3.3V for the STM32 PA1 ADC pin

V_PA1 = 5V × 82k/(33k+82k) = 3.57V — slightly high, but PA1 is 5V tolerant on F405

Allows firmware to detect USB connection state and switch between USB and UART modes

Design follows ST Application Note AN4879 for USB-C on STM32

C15 = 100 nF decouples VBUS at the sense point

D+/D– Routing:

USB 2.0 differential pair routed as matched-length traces

Both sides of the USB-C connector (A6/B6 for D+, A7/B7 for D–) tied together per USB-C spec

SBU1 and SBU2 left as no-connect — not used for USB 2.0

6. Inertial Measurement Unit — ICM-42688-P

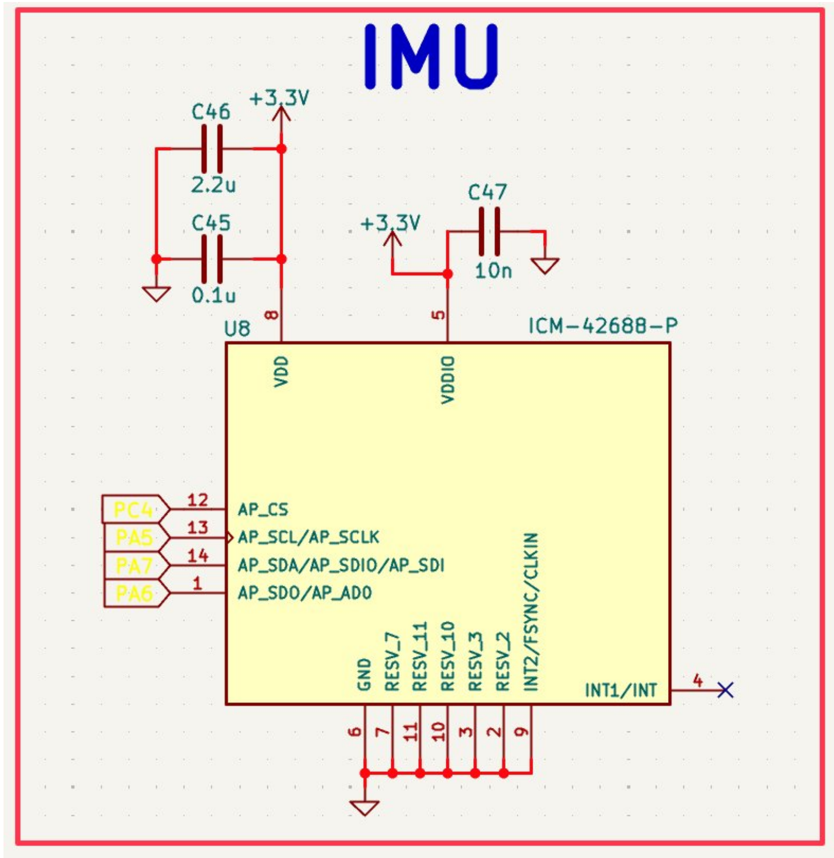

Figure 9: ICM-42688-P IMU schematic — SPI1 interface, VDD/VDDIO decoupling

The TDK InvenSense ICM-42688-P is a 6-axis IMU (3-axis gyroscope + 3-axis accelerometer) selected for its industry-leading noise density (gyro: 2.8 mdps/√Hz), 32 kHz ODR capability, and native SPI support up to 24 MHz. It is the current state-of-the-art IMU used in high-performance flight controllers, replacing the older MPU-6000/ICM-20689 series. Its performance directly determines flight controller loop quality.

Interface — SPI1 (PA5/PA6/PA7, CS: PC4):

PA5 = SPI1_SCK (SCLK) — clock up to 24 MHz in SPI mode 3

PA6 = SPI1_MISO (AP_SDO) — data from IMU to MCU

PA7 = SPI1_MOSI (AP_SDA/AP_SDIO) — data from MCU to IMU

PC4 = dedicated chip select (active low) — GPIO controlled by firmware

SPI chosen over I2C for speed: 24 MHz SPI vs 400 kHz I2C max

At 24 MHz SPI, a full 6-axis read (12 bytes) completes in ~5 µs — critical for 8 kHz PID loops

Power Supply Decoupling:

VDD pin (pin 8): C46 = 2.2 µF + C45 = 0.1 µF in parallel

2.2 µF handles bulk charge storage; 0.1 µF targets high-frequency noise

VDDIO pin (pin 5): C47 = 10 nF — IO logic supply, separate decoupling per datasheet

Both caps placed as close as possible to respective pins (<1 mm)

Separate VDD and VDDIO decoupling prevents digital switching on IO pins from coupling into the sensitive analog measurement core

RESV Pins:

- All RESV (reserved) pins connected to GND per ICM-42688-P datasheet requirement
- Floating reserved pins can cause undefined behavior or increased noise floor
- INT1/INT pin left as no-connect — firmware polls via SPI rather than using interrupt

7. Barometer — BMP388

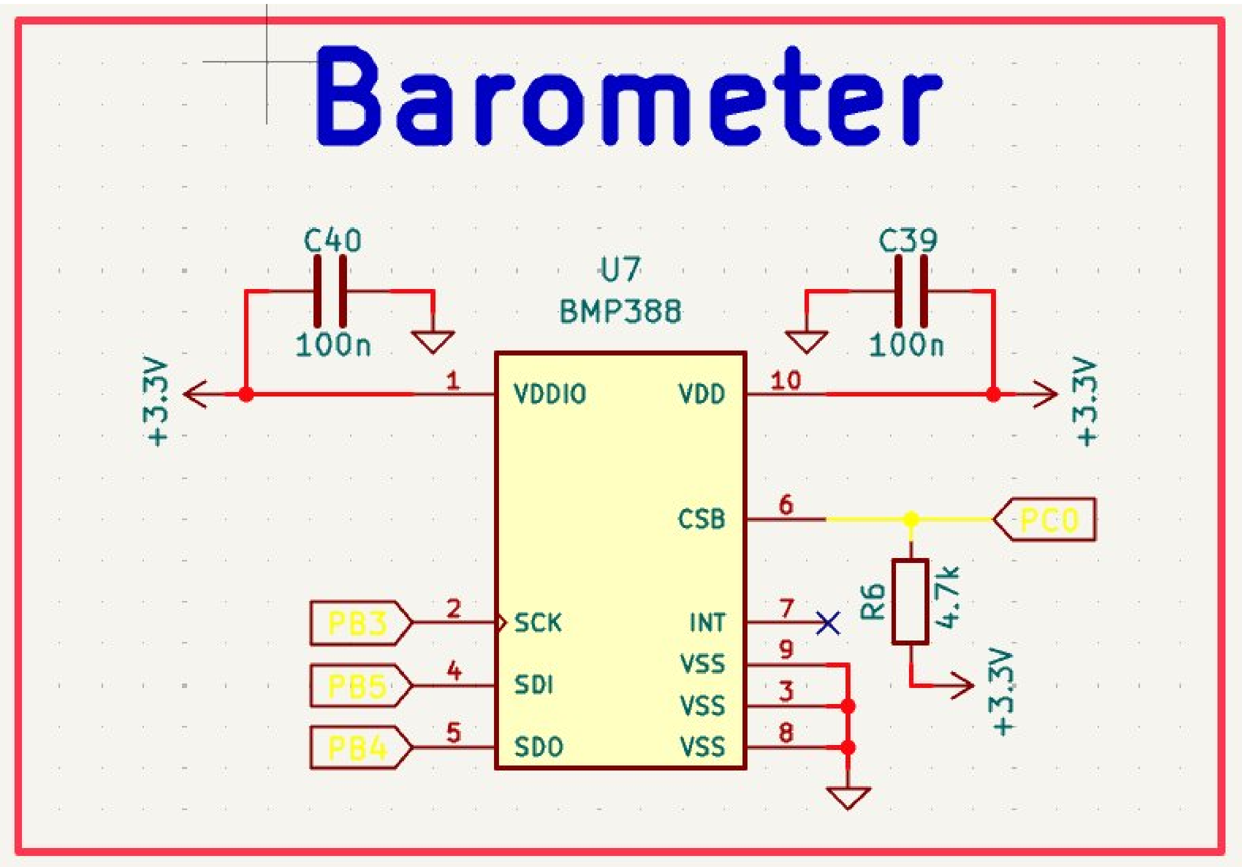

Figure 10: BMP388 barometer schematic — SPI3 interface, separate VDDIO/VDD decoupling

The Bosch BMP388 is a high-precision barometric pressure and temperature sensor used for altitude hold and variometer functions in Betaflight. It communicates via SPI3 and is powered from the 3.3V rail. The BMP388 was chosen over the older BMP280 for its improved noise performance (±0.08 hPa RMS) and lower current consumption.

Interface — SPI3 (PB3/PB4/PB5, CS: PC0):

PB3 = SPI3_SCK, PB4 = SPI3_MISO (SDO), PB5 = SPI3_MOSI (SDI)

PC0 = chip select, with R6 (4.7 kΩ) pull-up to 3.3V

Pull-up on CS ensures the barometer is deselected (CS high) during power-up, preventing spurious SPI transactions before firmware initializes the bus

CSB must be pulled high within the BMP388 to select SPI mode over I2C

Decoupling — C39 (VDD), C40 (VDDIO) = 100 nF each:

BMP388 has separate VDD (power) and VDDIO (IO logic) supply pins

Independent decoupling on each pin prevents logic switching on VDDIO from disturbing the pressure measurement reference on VDD

100 nF value per BMP388 datasheet recommendation

INT pin left as no-connect — Betaflight driver polls the barometer at a fixed rate

VSS pins (3, 8, 9) all connected to GND pour

8. OSD — AT7456E

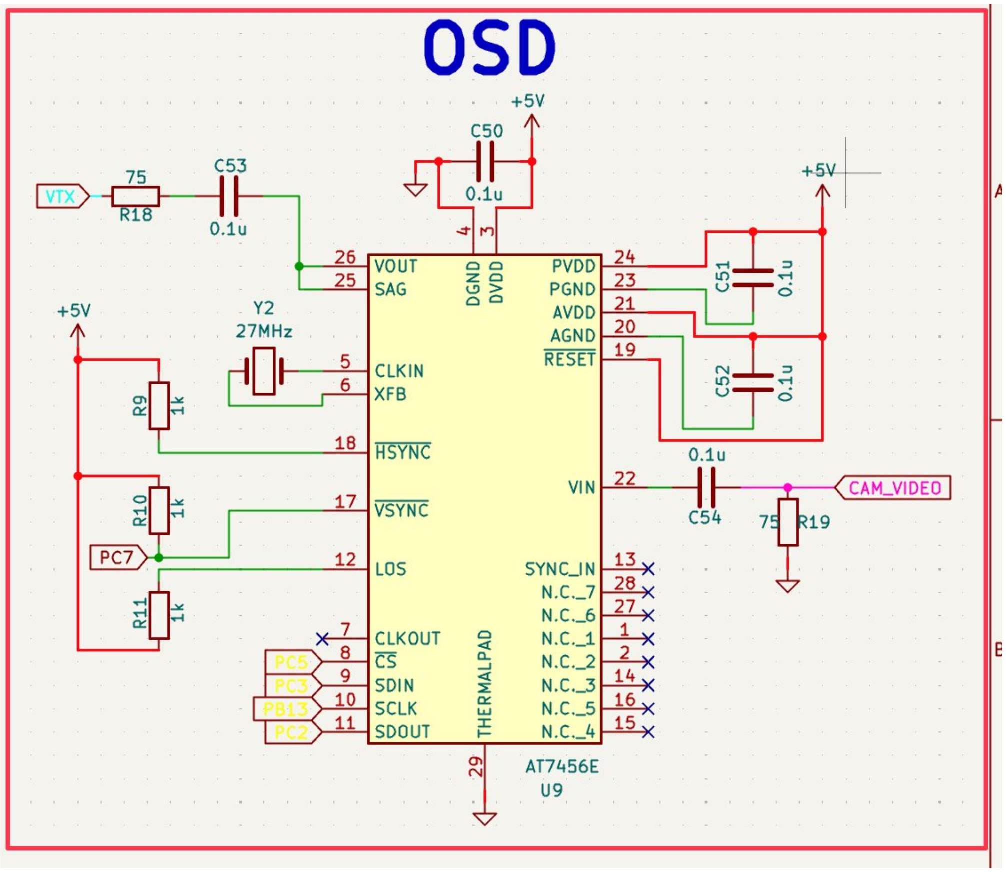

Figure 11: AT7456E OSD schematic — SPI2, 27 MHz crystal, video I/O with 75Ω termination

The ZHONGKEWEI AT7456E is an on-screen display (OSD) chip that overlays flight telemetry (speed, altitude, battery voltage, RSSI) onto the FPV camera video feed before transmission to the video transmitter. It operates on 5V (powered from the buck 5V rail) and communicates with the STM32 via SPI2.

Interface — SPI2 (PC3/PB13/PC2, CS: PC5):

PC3 = SPI2_MOSI (SDIN), PB13 = SPI2_SCK (SCLK), PC2 = SPI2_MISO (SDOUT)

PC5 = chip select, PC7 = LOS (Loss of Signal output from OSD)

PC7 monitors the OSD LOS pin — firmware can detect loss of camera sync

HSYNC (pin 18) and VSYNC (pin 17) connected to PC7 area for sync detection

27 MHz Crystal — Y2:

AT7456E requires an external 27 MHz crystal for its internal video clock PLL

27 MHz is the standard clock for PAL/NTSC video timing generation

Crystal connected to CLKIN (pin 5) and XFB (pin 6) — XFB is the crystal feedback pin

R9, R10, R11 (1 kΩ each) form the bias network for the crystal oscillator amplifier

Video Input — VIN (pin 22):

CAM_VIDEO net connects the FPV camera output to the OSD VIN pin

C54 = 0.1 µF AC coupling capacitor — removes DC bias from camera signal

R19 = 75 Ω termination to GND — matches video impedance standard (75 Ω)

Without 75 Ω termination, video signal reflections cause ghosting artifacts

Video Output — VOUT (pin 26):

R18 = 75 Ω series termination on VOUT to VTX connector

C53 = 0.1 µF AC coupling on VTX signal

SAG pin (25) connected — monitors sync amplitude for AGC

Power Supply — PVDD, AVDD (5V):

C50, C51, C52 = 0.1 µF decoupling on PVDD, AVDD, and DGND/DVDD pins

AT7456E requires separate decoupling on analog and digital supply domains

RESET pin tied high (to 5V) through resistor network — chip active on power-up

9. Magnetometer — QMC5883L

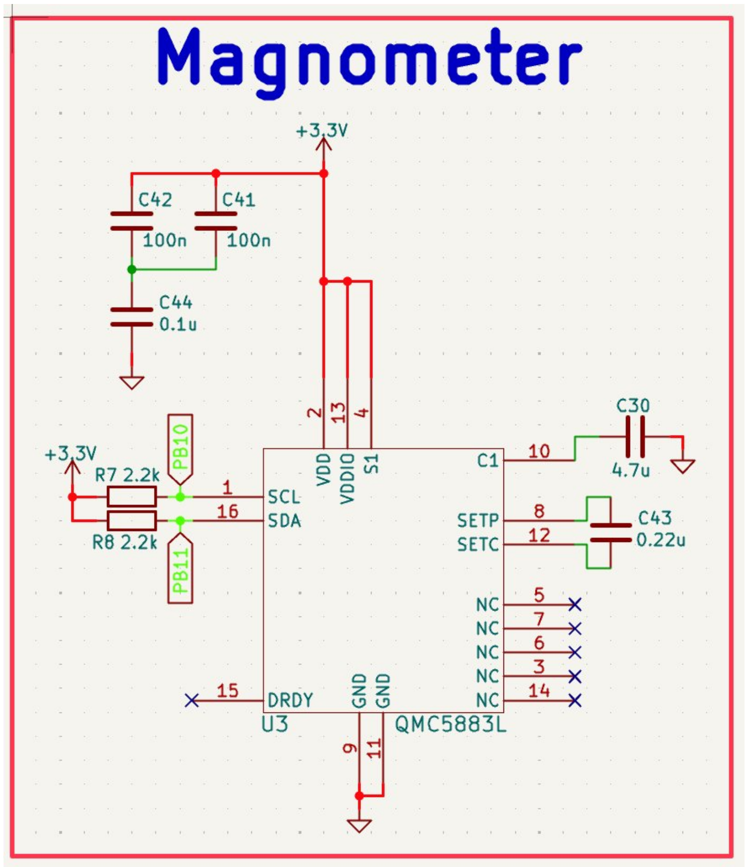

Figure 12: QMC5883L magnetometer schematic — I2C2 with pull-ups, SET/RESET capacitors

The QST QMC5883L is a 3-axis magnetic sensor providing heading information for GPS navigation and compass hold. It operates on I2C2 and uses an HMC5883L-compatible register set, allowing direct compatibility with existing Betaflight magnetometer drivers.

Interface — I2C2 (PB10/PB11):

PB10 = I2C2_SCL, PB11 = I2C2_SDA

R7 = R8 = 2.2 kΩ pull-up resistors to 3.3V on SCL and SDA

Pull-up value selection: I2C bus capacitance ~50 pF, f_SCL = 400 kHz (Fast mode)

Time constant RC = 2.2 kΩ × 50 pF = 110 ns — safely below the 300 ns Fast mode limit

Lower values (1 kΩ) would increase current draw; higher (4.7 kΩ) would slow edge rates

SET/RESET Circuit — C43 (0.22 µF) between SETP/SETC:

The QMC5883L requires periodic SET/RESET pulses to demagnetize internal sense elements

C43 = 0.22 µF connected between SETP (pin 8) and SETC (pin 12)

The IC drives a current pulse through C43 to generate the magnetic reset pulse

Value from QMC5883L application note — must not be omitted or compass calibration drifts

Decoupling:

C41, C42 = 100 nF on VDD and VDDIO

C44 = 0.1 µF additional bypass on VDD

C30 = 4.7 µF bulk cap on C1 pin (internal charge pump output)

S1 pin connected to VDD per datasheet — enables I2C mode

Placement Consideration:

The QMC5883L measures Earth's magnetic field, which is very weak (~25–65 µT). Any nearby current-carrying traces create magnetic fields that dwarf the Earth's field, corrupting compass readings. For this reason the magnetometer should be mounted as far as possible from high-current paths — ideally on a mast or at the board edge away from motor power traces, ESC wiring, and the power section. In this design it is placed at the far corner of the board from the power section, and in a production implementation would typically be on an external module connected via the I2C header.

10. GPS — SAM-M10Q-00B

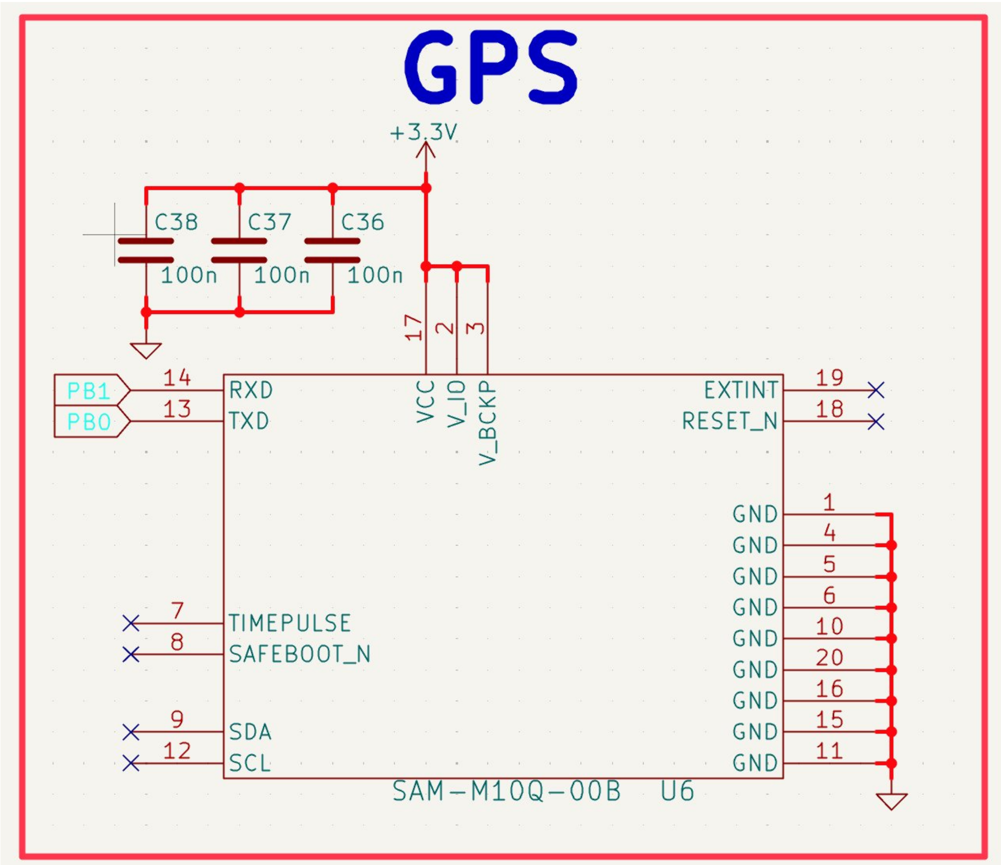

Figure 13: SAM-M10Q GPS module schematic — UART3, three-rail decoupling

The u-blox SAM-M10Q is a high-performance GNSS module in a compact 3×3 mm LCC package with an integrated patch antenna. It supports GPS, GLONASS, Galileo, and BeiDou simultaneously, providing position accuracy of ~1.5 m CEP. The module is mounted directly on the PCB rather than using an external GPS connected via JST cable, reducing wire inductance and weight.

Interface — UART3 (PB0/PB1):

PB0 = USART3_TX → GPS RXD (pin 14)

PB1 = USART3_RX → GPS TXD (pin 13)

Default baud rate: 9600 bps, upgradeable to 115200 via UBX configuration messages

Simple UART — no hardware flow control required for GPS data rates

Power Supply — Three Independent Rails:

VCC (pin 17) = 3.3V: main module supply — C36 = 100 nF decoupling

V_IO (pin 2) = 3.3V: IO logic supply — C37 = 100 nF decoupling

V_BCKP (pin 3) = 3.3V: backup supply for RAM/RTC — C38 = 100 nF decoupling

Each rail has independent decoupling to prevent cross-coupling between domains

V_BCKP keeps the GPS hot-start data alive — significantly reduces TTFF on power cycle

Unused Pins:

TIMEPULSE (pin 7) and SAFEBOOT_N (pin 8) — left as no-connect

RESET_N (pin 18) — no-connect: internal pull-up keeps module running

EXTINT (pin 19) — no-connect: wake-on-movement not required

SDA/SCL (pins 9/12) — no-connect: I2C interface not used, UART selected

All GND pins (1, 4, 5, 6, 10, 11, 15, 16, 20) tied to GND plane

Placement Consideration:

The SAM-M10Q has an integrated patch antenna optimized to receive from directly above. It must be placed with a clear sky view — away from board edges that might block satellite signals, and with no metal components taller than the module within the antenna's field of view. The module is placed in the bottom-right corner of the board in a relatively unobstructed area.

11. FPV Camera Input

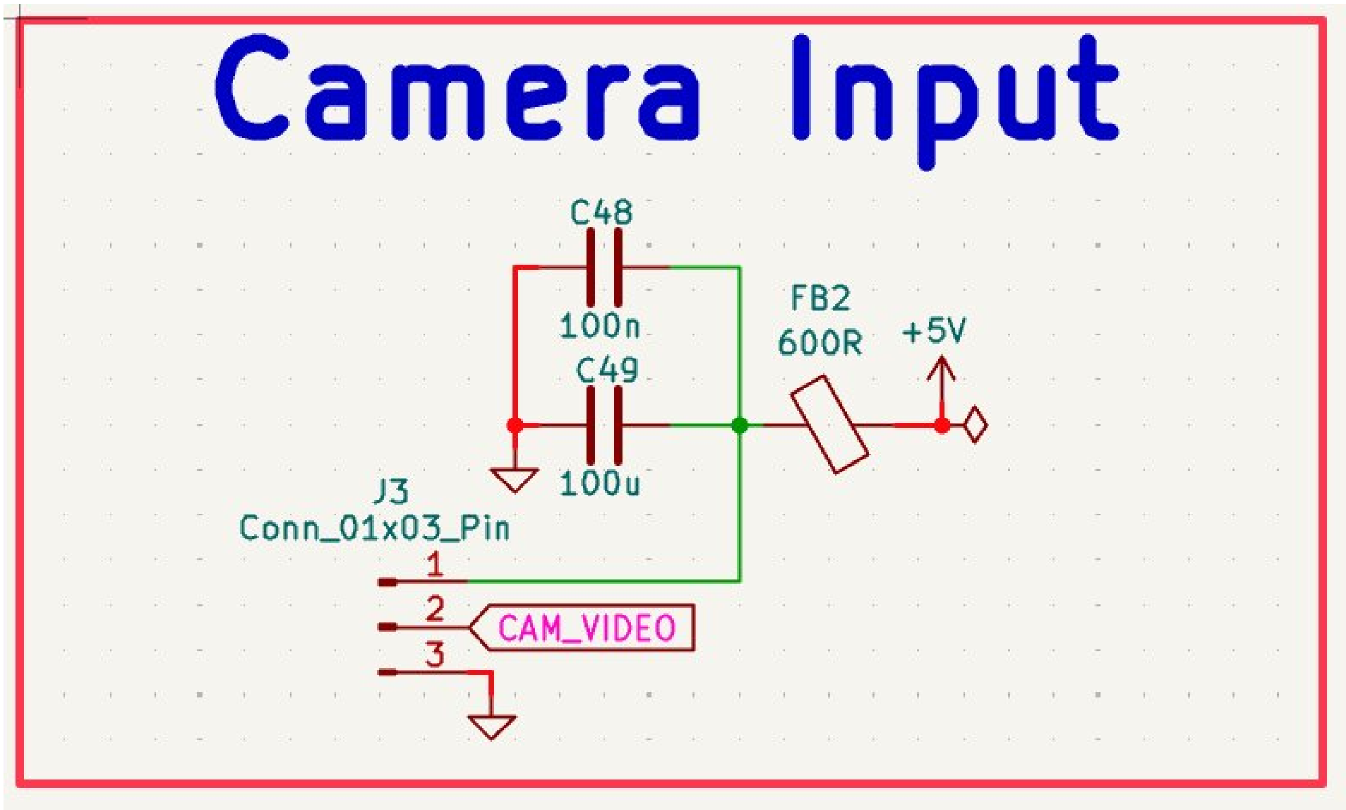

Figure 14: FPV camera input circuit — JST-SH 3-pin, 600Ω ferrite bead, bulk/bypass capacitors

The FPV camera input provides power and receives the analog video signal from the FPV camera. The 3-pin JST-SH connector (J3) carries +5V, video signal (CAM_VIDEO), and GND.

FB2 — 600 Ω @ 100 MHz Ferrite Bead:

The 5V camera power is filtered through a 600 Ω ferrite bead before the camera connector

FPV cameras contain switching regulators that generate conducted noise on the power rail

The ferrite bead prevents camera switching noise from propagating back into the 5V bus

600 Ω chosen for higher attenuation than the 120 Ω FB1 — camera noise is typically stronger

C49 — 100 µF 1206:

Large bulk capacitor on the camera power line after the ferrite bead

Acts as local charge reservoir — cameras draw sudden current spikes when driving video

Prevents voltage droop on the camera supply during high-current transients

1206 package for adequate voltage rating and low ESR

C48 — 100 nF:

High-frequency bypass in parallel with C49 — handles fast transients C49 is too slow for

Placed as close as possible to the connector power pin

CAM_VIDEO Net:

Video signal pin (pin 2) connects to CAM_VIDEO global net label

This net routes to the OSD VIN pin (AT7456E) via AC coupling capacitor C54

12. Buzzer Driver

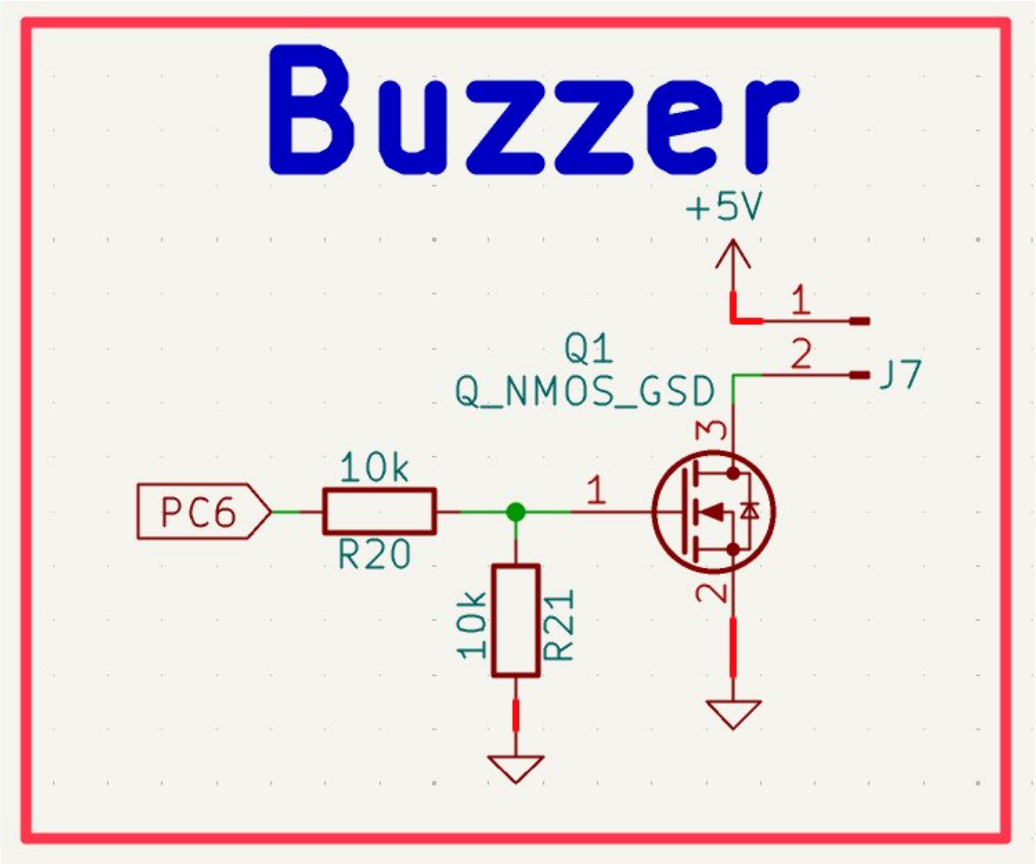

Figure 15: Buzzer driver circuit — NMOS Q1 with gate resistor/pull-down, 5V buzzer connector J7

The buzzer provides audible alerts for arming, disarming, low battery, and lost model beeping. Since the STM32 GPIO can only source ~25 mA and a buzzer typically requires 100–200 mA at 5V, a MOSFET switch is used to drive the buzzer from the 5V rail.

Q1 — N-channel MOSFET (SOT-23):

STM32 PC6 GPIO drives the MOSFET gate through R20 (10 kΩ series resistor)

R20 limits gate current and slows gate charge to reduce EMI from fast switching

R21 (10 kΩ pull-down to GND) ensures gate is held low when PC6 is high-impedance

Without R21, floating gate could cause buzzer to activate spuriously during boot

When PC6 goes high: MOSFET turns on, connects buzzer GND to system GND, buzzer sounds

Buzzer connected between J7 and +5V — MOSFET switches the low side (GND)

Low-side switching is preferred: MOSFET source connected to GND, V_GS = V_PC6 = 3.3V

No bootstrap circuit needed — straightforward enhancement-mode NMOS operation

13. Connectors & Peripheral I/O

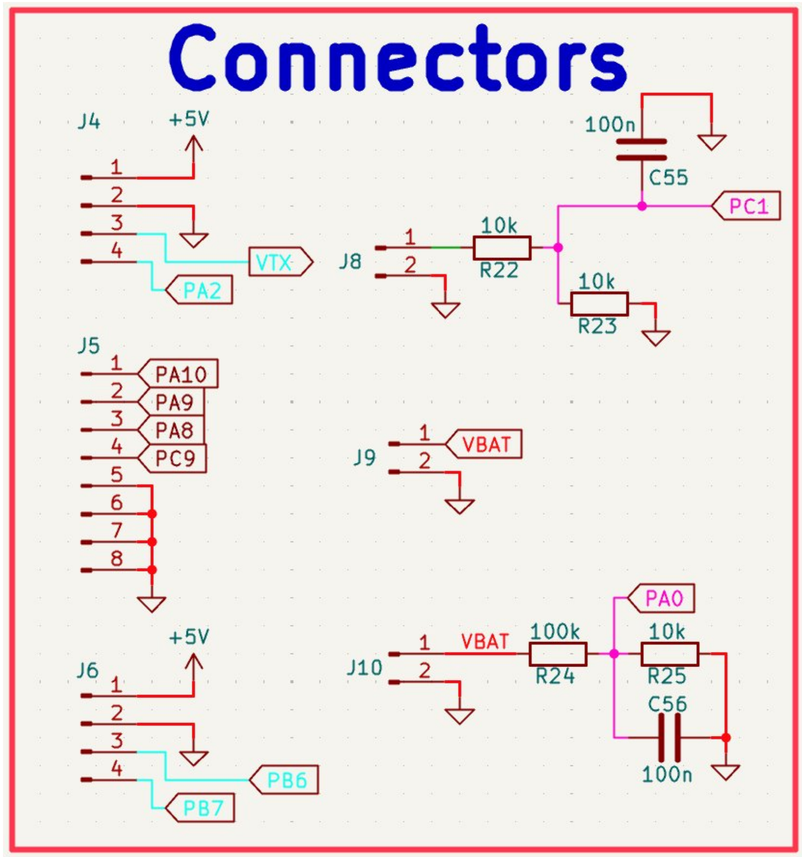

Figure 16: Peripheral connectors — motor ESC outputs (J4/J5/J6/J7), VTX, receiver, battery sense

All peripheral connectors use JST-SH 1.0 mm pitch (BM-series) for signal connections and XT30 for power. JST-SH was chosen for its small size (critical on a compact FC), positive retention locking, and industry standard use in FPV hardware.

Motor ESC Outputs — J5 (8-pin JST-SH):

PA10 (TIM1_CH3), PA9 (TIM1_CH2), PA8 (TIM1_CH1) = Motor 1/2/3 DShot outputs

PC9 (TIM8_CH4) = Motor 4 DShot output

Remaining pins = GND and +5V for ESC signal reference and BEC

DShot is a digital motor protocol — eliminates analog PWM calibration, provides error checking, and supports bidirectional telemetry

Receiver Input — J5 (4-pin JST-SH, separate):

PA10/PA9 = USART1 RX/TX for CRSF/ELRS ExpressLRS receiver protocol

+5V and GND for receiver power — most modern receivers run on 5V

VTX — J4 (4-pin JST-SH):

PA2 = USART2_TX for SmartAudio VTX control

+5V and GND for VTX power

SmartAudio allows Betaflight to control VTX power level, channel, and pit mode

Current Sensor — J8 (2-pin):

PC1 = ADC input for current sensor analog output

R22/R23 (10 kΩ) voltage divider scales sensor output to ADC range

C55 = 100 nF anti-aliasing filter on ADC input

Allows Betaflight to display real-time current draw and mAh consumed

Battery Voltage Sense — J10/R24/R25:

J10 = XT30 battery connector (also main power input)

R24 = 100 kΩ (upper), R25 = 10 kΩ (lower) voltage divider

Scales VBAT (up to 25.2V) to PA0 ADC: V_PA0 = 25.2 × 10k/(100k+10k) = 2.29V — within 3.3V ADC range

C56 = 100 nF anti-aliasing filter

Allows Betaflight to display cell voltage and trigger low battery warnings

ESC Power — J9 (XT30):

Passes VBAT directly to ESCs — no filtering (ESCs have their own capacitors)

XT30 rated for 30A continuous — adequate for 4× brushless motors

14. PCB Layout

14.1 Stackup & Layer Strategy

The board uses a 4-layer JLCPCB standard stackup: F.Cu / In1.Cu / In2.Cu / B.Cu, with 1 oz copper on outer layers and 0.5 oz on inner layers (upgraded to 1 oz for this design). Total board thickness: 1.6 mm. Board dimensions: 74 × 108 mm.

Layer	Function	Contents
F.Cu (Top)	Signal + Component	Component pads, critical short traces, power traces from vias
In1.Cu	Ground Plane	Solid unbroken GND — reference plane for all signals
In2.Cu	Signal Overflow	Secondary signal routing when F.Cu is congested
B.Cu (Bottom)	3.3V Distribution	3.3V power backbone, bottom-side pads, ground pour

The key architectural decision was to use B.Cu as the 3.3V distribution backbone. 3.3V is distributed across the bottom copper layer as a wide pour, with vias popping up to F.Cu pads at each component. This keeps the top layer clean for signal routing and maintains a near-continuous ground plane on In1.Cu directly above all F.Cu signals. In2.Cu handles signal overflow — UART traces, SWD, and other low-priority signals that cannot be routed on F.Cu without excessive congestion.

The ground plane on In1.Cu is intentionally kept as a solid copper pour with no signal routes cutting through it. Any signal that must cross layers does so via a via, and a ground via is placed as close as possible to the signal via to provide a local return path. Without this, the return current must travel a longer path around the signal via, creating a loop antenna that radiates EMI.

14.2 Overall Placement Strategy

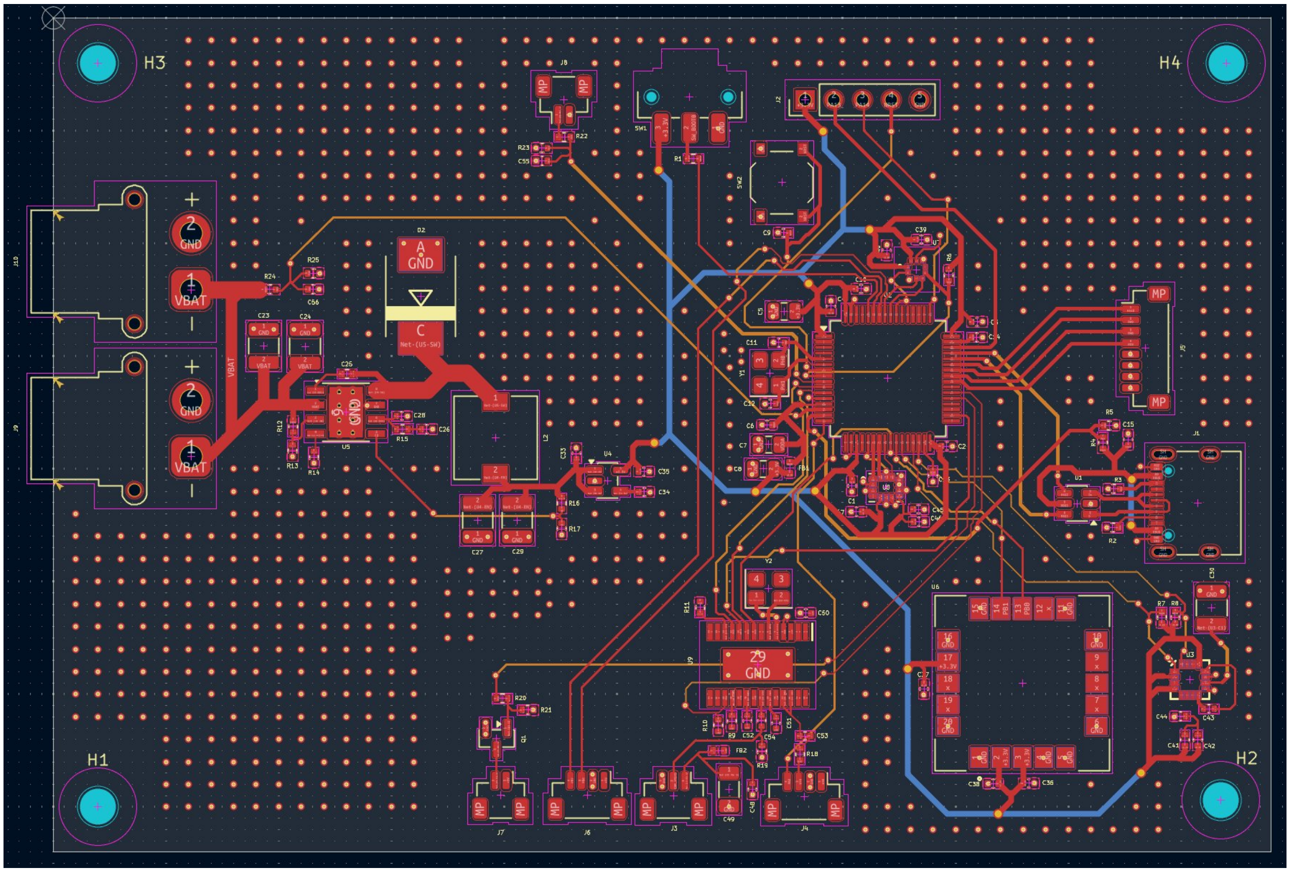

Figure 17: Full PCB top view — component placement overview

Component placement follows a functional grouping strategy, with high-power and noisy circuits physically separated from sensitive analog and high-frequency digital circuits. The power section (buck converter, inductor, diode, bulk capacitors) occupies the left side of the board, as far as possible from the IMU and barometer. The STM32 is centrally located with its decoupling caps immediately adjacent. The magnetometer is at the far bottom-right corner — maximum distance from all high-current paths. The GPS module is at the bottom-center, positioned for clear sky view. Motor output connectors are along the bottom edge for clean wiring to ESCs.

Four M3 mounting holes (H1–H4) at the corners are placed outside all routing areas and provide mechanical attachment to the drone frame. The 74×108 mm board size was chosen to match common 30.5×30.5 mm and 20×20 mm FC mounting patterns when centered.

14.3 Power Section Layout

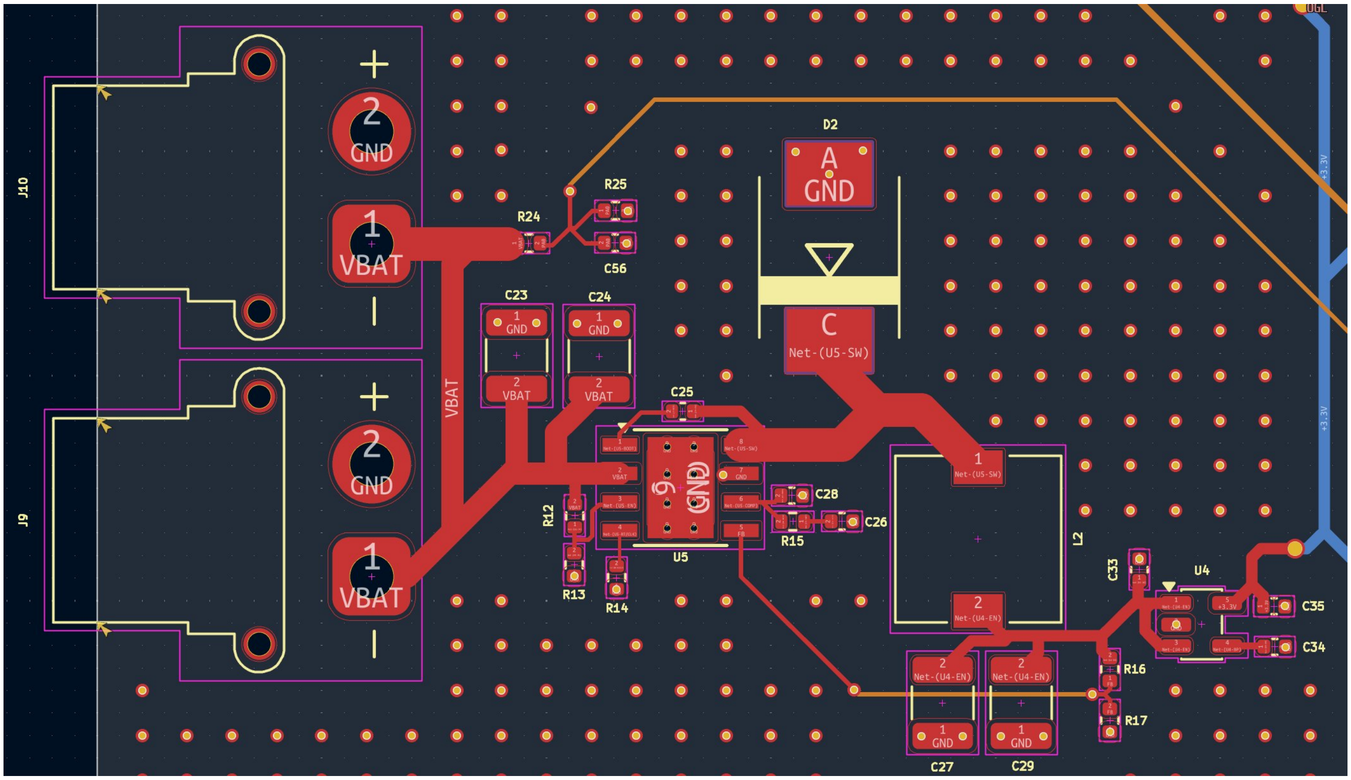

Figure 18: Power section PCB layout — TPS54360, inductor L2, diode D2, bulk capacitors

The power section layout is critical for EMI and efficiency. The TPS54360DDA (U5) is placed with the input capacitors (C23, C24) immediately to its left at the VIN pin, and the output capacitors (C27, C29) to its right after the inductor. This minimizes the area of both the input and output current loops.

Switched Node (SW — Net-(U5-SW)):

The SW node trace (between pin 8, D2, and L2) is kept as short and narrow as possible

No copper poured over or near the SW node — this trace switches at 600 kHz with 25V swings, radiating EMI proportional to its area

D2 catch diode placed as close to TPS54360 SW pin as physically possible

Ground return from D2 is a short, direct trace to the GNDPAD thermal via array

GNDPAD Thermal Vias:

The TPS54360DDA has an exposed thermal pad (pin 9) on the bottom of the IC

Array of thermal vias connect this pad to the In1.Cu ground plane for heat dissipation

Without thermal vias, the pad temperature would rise above the 150°C rating under load

Input Capacitors — C23, C24 (4.7 µF, 1210):

Placed immediately adjacent to VIN pin — shortest possible trace

Ground pins of caps connect directly to GND pour with via to In1.Cu

Minimizes input loop inductance, reducing voltage spike on VIN during switching

Output Capacitors — C27, C29 (47 µF, 1210):

Placed at the output end of the inductor, between inductor output and LDO input

Both caps share a GND via cluster to minimize loop impedance

LDO — U4 (SPX3819, SOT-23-5):

Placed immediately to the right of the output capacitors

C33 (input cap) at IN pin, C34+C35 (output caps) at OUT pin

Short traces throughout — LDO output drives the global 3.3V B.Cu distribution plane

14.4 STM32 Area Layout

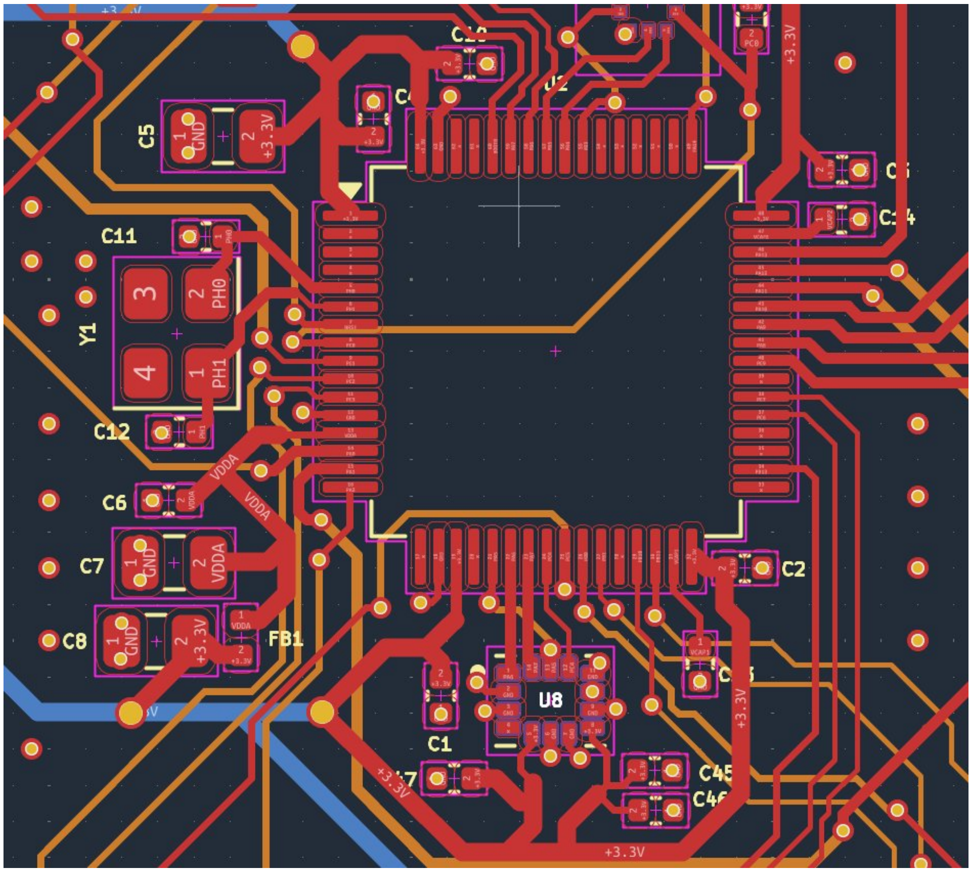

Figure 19: STM32F405 area — decoupling capacitors on all VDD pins, crystal C11/C12, VDDA filter

The STM32F405 LQFP-64 is placed in the upper-center of the board. Each of the four VDD pins has its own 100 nF decoupling capacitor placed as close as physically possible — within 1 mm of the pin, with the cap GND pad connected by a very short trace to a via down to the In1.Cu GND plane. The 4.7 µF bulk cap C5 is placed just outside the LQFP body.

The VCAP capacitors C13 and C14 (2.2 µF each) are placed at VCAP1 (pin 31) and VCAP2 (pin 47) with minimal trace length. These are on the left and right sides of the STM32 respectively, matching pin positions. Long VCAP traces add inductance that destabilizes the internal core voltage regulator.

The crystal Y1 is placed to the left of the STM32, directly adjacent to PH0 and PH1 pins. C11 and C12 load capacitors are placed within 2 mm of the crystal body, with their GND pins connected to a dedicated GND via rather than the general GND pour — this provides a clean, low-impedance return path that minimizes noise coupling into the crystal circuit. The crystal traces are kept short (<5 mm total) and routed away from any high-speed signals.

14.5 IMU Layout — U8

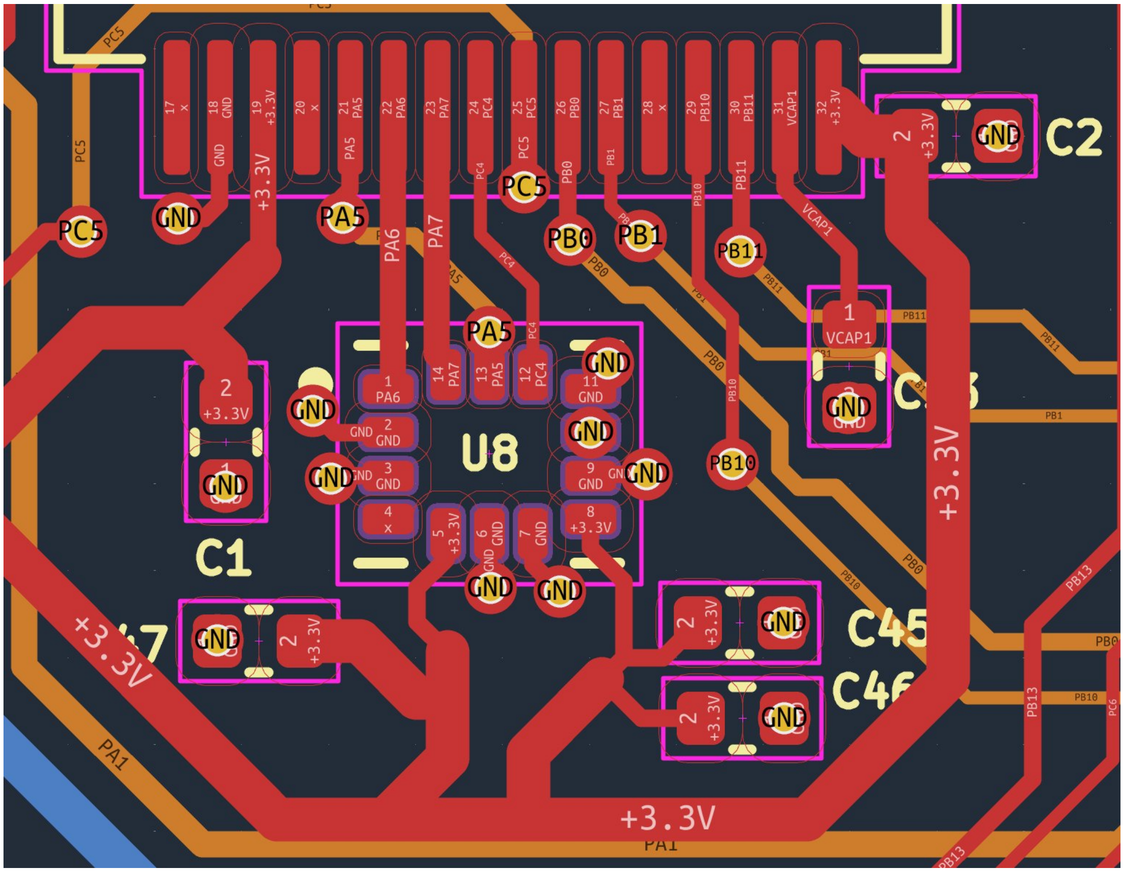

Figure 20: ICM-42688-P IMU area — QFN package, ground via stitching, decoupling caps C1/C45/C46

The ICM-42688-P is a 14-pin QFN package (3×2.5 mm) with a central thermal/GND pad. The thermal pad is connected to the In1.Cu GND plane via multiple thermal vias, providing both electrical ground connection and mechanical adhesion.

- C45 (0.1 µF) and C46 (2.2 µF) placed on VDD pin (pin 8) within 1 mm
- C1 (additional 100 nF) on adjacent VDD trace for distributed bypass
- C47 (10 nF) on VDDIO pin (pin 5) — placed on opposite side of IC
- All SPI traces (PA5/6/7, PC4) routed on F.Cu with ground pour directly above on In1.Cu
- Ground vias placed adjacent to each SPI signal via to minimize ground return loop

The IMU should be placed as close to the center of the drone as possible to minimize the vibration arm that amplifies mechanical resonance. On this board it is placed near the geometric center, and soft-mount grommets are used at the mounting holes to further isolate vibration.

14.6 OSD & 5V Area Layout — U9, Y2

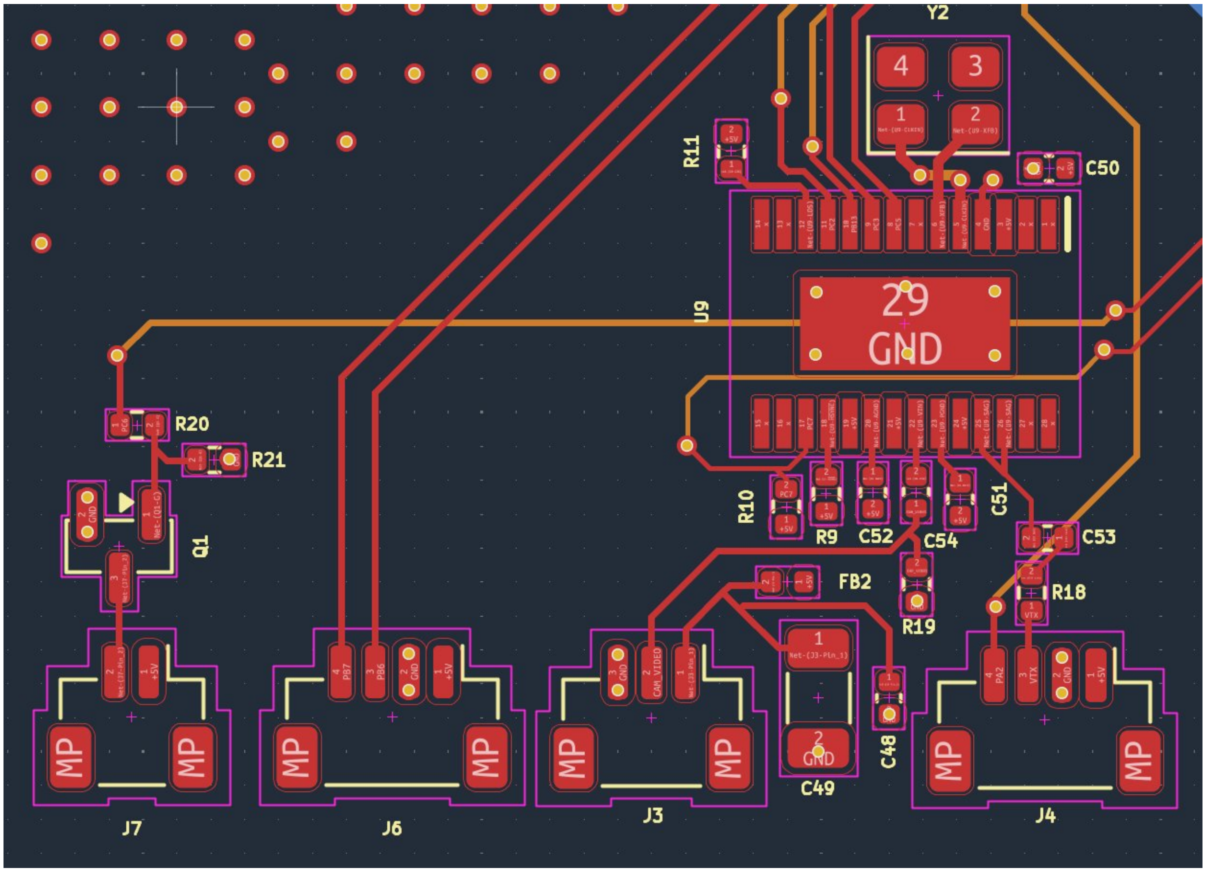

Figure 21: AT7456E OSD and 5V area — Y2 27MHz crystal, decoupling, video routing, R18/R19 termination

The AT7456E OSD chip (U9, 29-pin SOIC) occupies the lower-center of the board. The 27 MHz crystal Y2 is placed immediately adjacent to the CLKIN and XFB pins, with the shortest possible traces. Decoupling capacitors C50, C51, C52 are placed at each power pin of the IC.

- R18 (75 Ω) and R19 (75 Ω) are the video termination resistors — placed close to connector
- Video traces (CAM_VIDEO, VTX) are kept short and away from SPI/UART digital signals
- The OSD thermal pad is connected to GND plane via vias
- FB2 (600 Ω ferrite) and C49 (100 µF) filter camera 5V supply at J3 connector
- Q1 buzzer MOSFET is placed next to J7 connector, with R20/R21 immediately at gate pin

14.7 Barometer Layout — U7

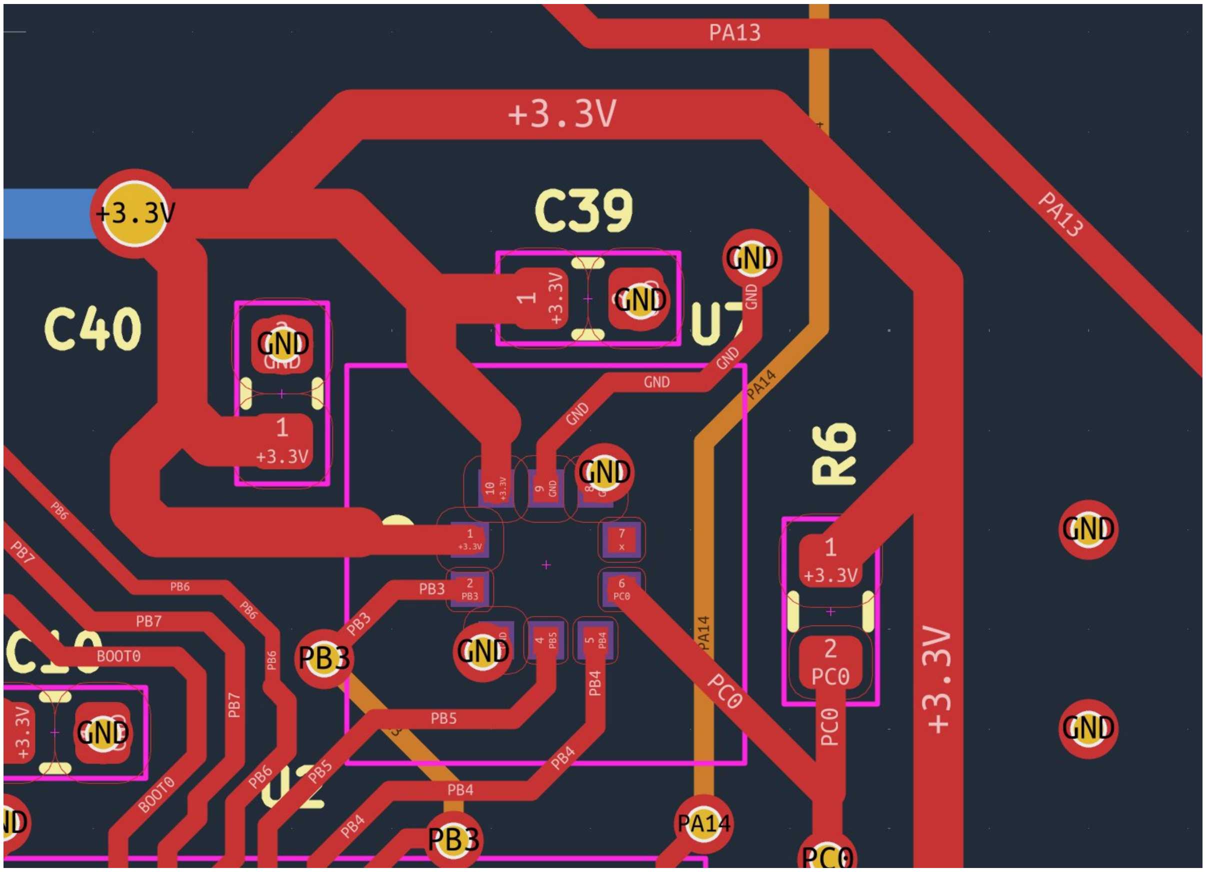

Figure 22: BMP388 barometer area — LCC package, VDDIO/VDD decoupling caps, SPI traces

The BMP388 is a small LCC package placed in the upper-right area of the board. C39 and C40 (100 nF each on VDD and VDDIO) are placed within 1 mm of their respective pins. The CS pull-up resistor R6 (4.7 kΩ) is placed adjacent to the CS pin. SPI3 traces (PB3/PB4/PB5) route from the STM32 to U7 on F.Cu.

The barometer measures absolute pressure to derive altitude. It should be placed away from heat-generating components (buck converter, power MOSFETs) because temperature gradients cause pressure measurement errors. It is also given vent holes in any enclosure to allow air pressure equalization.

14.8 Magnetometer Layout — U3

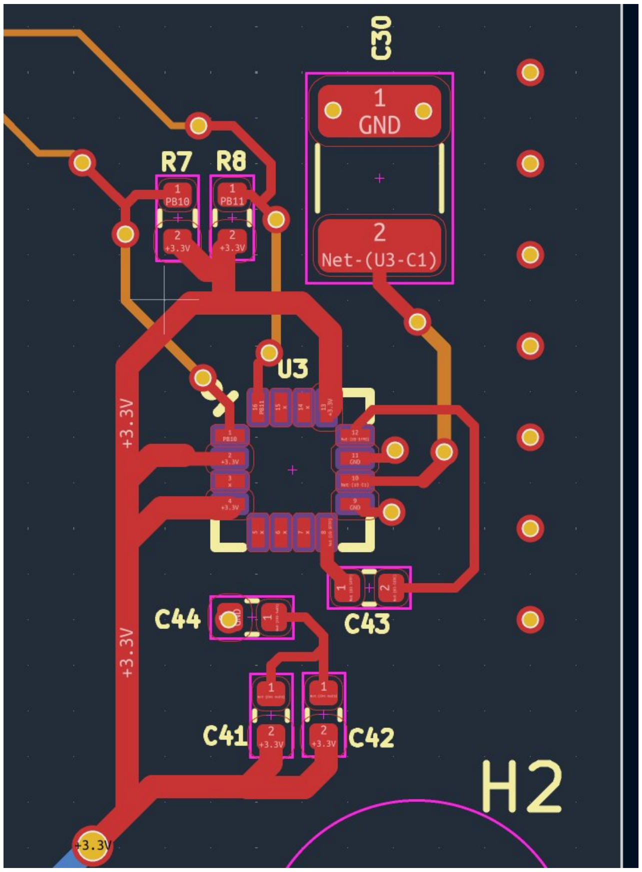

Figure 23: QMC5883L magnetometer layout — bottom-right corner, I2C pull-ups, SET/RESET cap C43

The QMC5883L (U3, LCC-16 package) is placed in the bottom-right corner of the board — the maximum distance from the power section and motor connectors on the opposite side. I2C pull-up resistors R7 and R8 (2.2 kΩ) are placed immediately adjacent to the SCL/SDA pins. C43 (0.22 µF SET/RESET cap) is placed between SETP and SETC pins with minimal trace length. C41, C42, C44 decoupling caps are placed within 2 mm of VDD pins.

Even with careful board placement, the magnetometer will experience interference from motor currents in flight. Betaflight's compass calibration routine and flight-time motor interference compensation algorithms are used in firmware to correct for this.

14.9 GPS Module Layout — U6

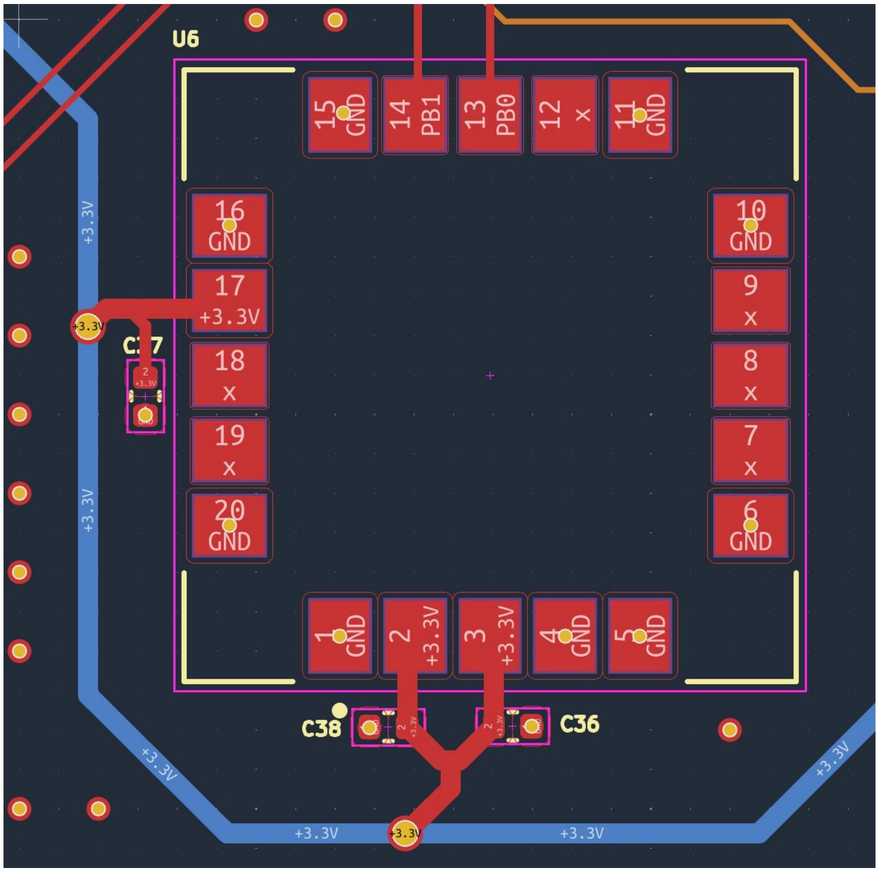

Figure 24: SAM-M10Q GPS module layout — LCC package, three decoupling rails, UART routing

The SAM-M10Q (U6) is a large LCC package (3×3 mm with exposed pads). Three sets of decoupling capacitors (C36, C37, C38 — 100 nF each) are placed immediately at the VCC, V_IO, and V_BCKP pins respectively. UART3 traces (PB0/PB1) route from the STM32 area to the GPS module pins. All GND pads of the module are connected to the In1.Cu GND plane via vias.

The GPS is placed with the antenna-side of the module (the die-integrated patch) facing upward with clear vertical sky view. No tall components are placed within the module's antenna footprint area. The blue traces visible in the layout image are the 3.3V B.Cu distribution layer showing through from the back copper.

14.10 USB-C Layout

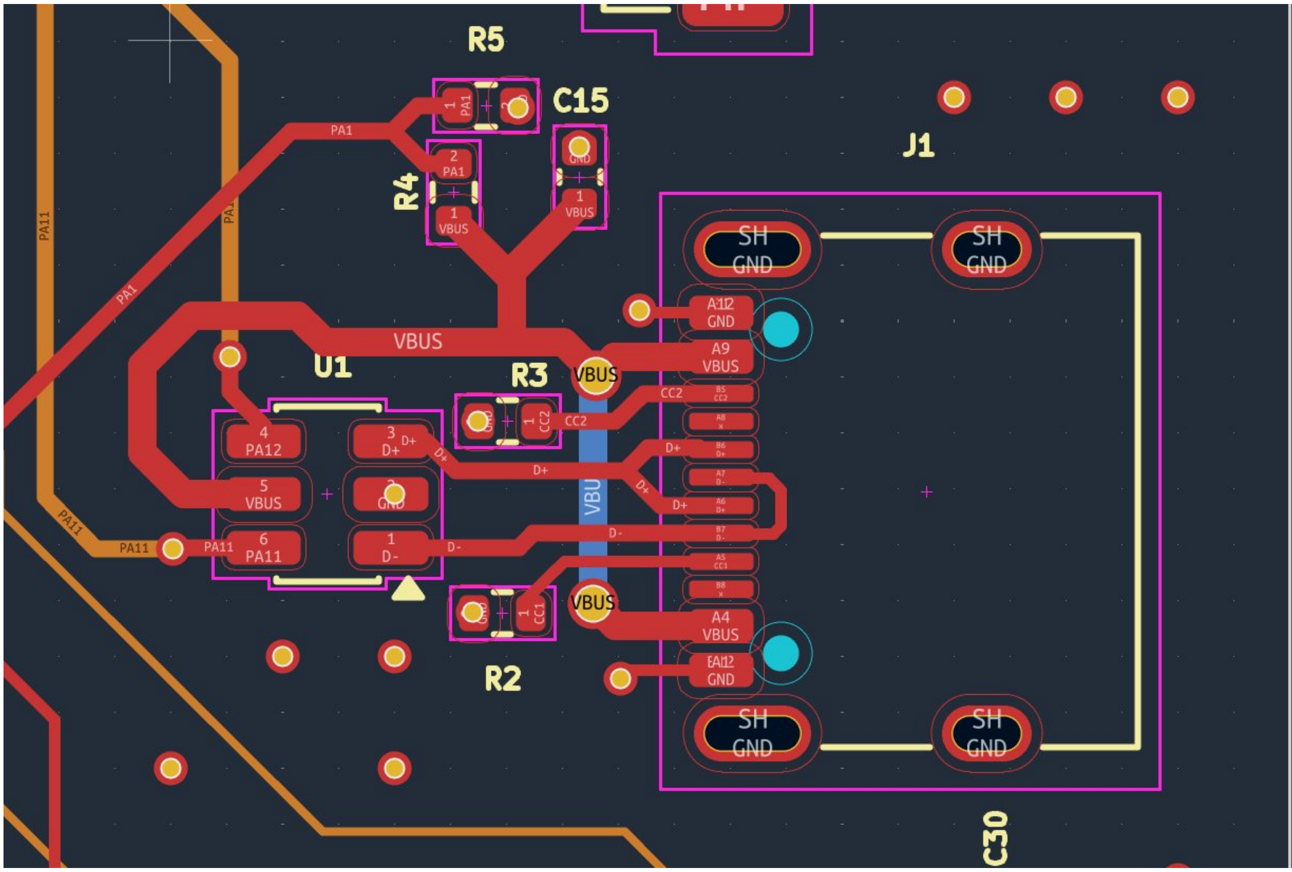

Figure 25: USB-C connector and ESD protection layout — J1, U1 USBLC6, CC resistors, VBUS divider

The USB-C connector (J1) is mounted at the right edge of the board for easy access. The USBLC6-2SC6 ESD protection IC (U1, SOT-23-6) is placed as close as possible to the connector — before any other routing of the D+ and D– lines. This ensures ESD events are clamped before they can travel along the PCB traces to the STM32.

- D+ and D– traces routed as a matched pair on F.Cu to U1, then to STM32 PA11/PA12
- CC1/CC2 pull-down resistors R2/R3 placed immediately at the connector CC pins
- VBUS divider R4/R5 placed adjacent to the VBUS pin with C15 bypass
- Shield and GND pins connected directly to GND pour with multiple vias

The blue differential pair visible in the PCB image shows the USB D+/D– traces routed together with consistent spacing. While full impedance-controlled 90 Ω differential routing is ideal for USB 2.0, at Full Speed (12 Mbps) the signal integrity requirements are relaxed enough that matched-length traces with consistent separation are sufficient.

14.11 Crystal & VDDA Area

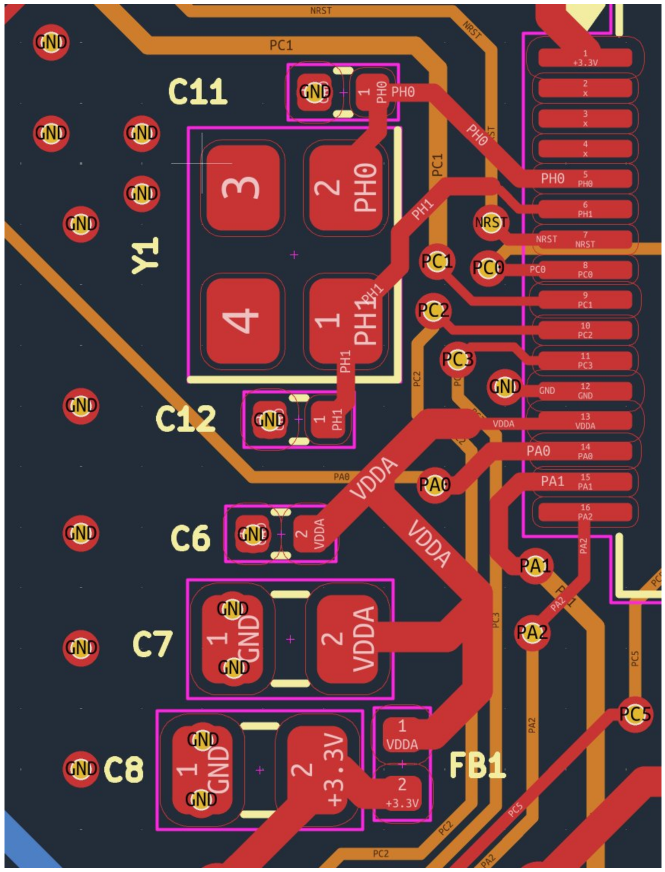

Figure 26: Crystal Y1 and VDDA filter area — C11/C12 load caps, FB1, C6/C7/C8 filter chain

The 8 MHz crystal Y1 is placed directly adjacent to STM32 pins PH0 and PH1 with load capacitors C11 and C12 flanking it. The traces from the crystal to the STM32 pins are kept under 5 mm total length. The crystal and its load caps are surrounded by a ground pour on F.Cu to shield the circuit from noise coupling.

The VDDA filter (FB1 ferrite bead, C6, C7, C8) is placed between the 3.3V B.Cu distribution layer and the VDDA pin of the STM32. C6 (10 nF) and C7 (1 µF) are on the input side of the ferrite bead, C8 (1 µF) is on the VDDA output side. The ferrite bead physically isolates the analog domain from the digital 3.3V rail noise.

14.12 Trace Widths & Impedance

Trace widths were assigned based on current capacity and signal type, using the IPC-2221 current capacity guidelines for 1 oz copper.

Net Class	Trace Width	Current / Use
VBAT Power	1.5 mm	Up to 5A battery input — wide to minimize resistive drop
5V Power	1.0 mm	Up to 3.5A buck output — wide power backbone
3.3V Power	0.5 mm	~500 mA digital rail — moderate width on B.Cu pour
GND Plane	Solid pour	Zero resistance — In1.Cu unbroken ground plane
SPI Signals	0.2 mm	Low current digital — narrow for routing density
I2C Signals	0.2 mm	Low current — standard signal trace
UART	0.2 mm	Low current digital signal

USB D+/D–	0.2 mm matched	Differential pair — matched length ±0.1 mm
Crystal	0.15 mm	Minimal length, surrounded by GND pour
ADC inputs	0.2 mm	Analog — kept away from switching signals

14.13 Ground Plane Strategy & Via Stitching

The In1.Cu ground plane is the most important layer in the stackup. It serves as the electromagnetic reference for every signal on the board, provides the return current path for all traces on F.Cu and In2.Cu, and acts as a thermal spreading layer.

- In1.Cu is a solid, unbroken copper pour — no signal traces cut through it
- Every component GND pad connects to In1.Cu via a short via within 1–2 mm
- Via stitching around the board perimeter connects F.Cu GND pour to In1.Cu at regular intervals
- Signal vias changing layers have a GND via placed <1 mm away — provides adjacent return path
- The IMU, barometer, and crystal areas have dense GND via stitching to create solid local ground islands

When a signal via transitions from F.Cu to In2.Cu, its return current must also transition between reference planes. Without a nearby GND via, the return current travels a long path around the via, forming a loop that radiates EMI. The nearby GND via provides a direct, low-inductance return path that keeps the loop area minimal.

15. Manufacturing & BOM

The board was manufactured by JLCPCB as a 4-layer PCB with SMT assembly. The BOM was prepared with LCSC part numbers for all SMT components, enabling JLCPCB's Economic assembly service for the bulk of the components. The following components are hand-soldered after board delivery:

Category	Detail
PCB Manufacturer	JLCPCB
Board Size	74 × 108 mm
Layers	4 (F.Cu / In1.Cu / In2.Cu / B.Cu)
Material	FR4 TG135
Thickness	1.6 mm
Copper Weight	1 oz outer / 1 oz inner
Surface Finish	LeadFree HASL
Min Trace/Space	0.15/0.15 mm
Min Via Drill	0.3 mm
Quantity	5 boards
SMT Assembly	JLCPCB Economic PCBA — top side
SMT Components	43 line items, all LCSC-sourced
Hand-Solder Parts	J1–J10 connectors, SW1, SW2, L2, Q1, Y1, Y2
Estimated Cost (bare)	~$44.92 for 5 boards
Assembly Cost	~$272 total for 5 assembled boards

The BOM contains components from Samsung, Uniroyal, TI, ST, Bosch, TDK InvenSense, u-blox, MaxLinear, Diodes Inc., ZHONGKEWEI, and QST — all verified against official datasheets. Non-standard E96 series resistors (feedback and UVLO networks) were sourced from LCSC extended parts inventory.